



自然语言处理

WELCOME!

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2025-2026-1 学期

基于 Julia, Junjie 等课件更新

面试问题

- 请简述传统RNN的核心缺陷是什么？LSTM 是如何通过自身结构设计解决这些缺陷的？
- LSTM 与 GRU 都属于改进型循环神经网络，两者的结构差异是什么？在实际应用中如何选择这两种模型？
- 在自然语言生成任务中，为什么直接用传统 RNN/LSTM 训练会出现“训练 - 推理分布偏移”问题？可以采用哪些方法缓解这一问题？长文本问题如何解决呢？

大纲

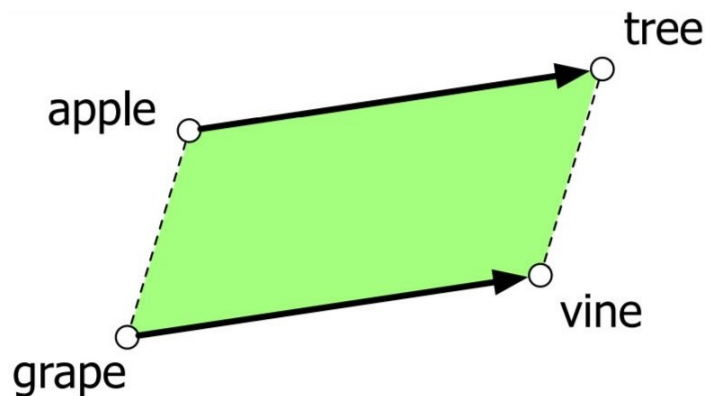
- 前言
- 循环神经网络
- 训练的常见问题
- 循环神经网络的使用
- 循环神经网络的理解
- 循环神经网络变种

类比关系

Analogical Relations

- 类比推理的经典模型是平行四边形模型 (Rumelhart and Abrahamson, 1973)

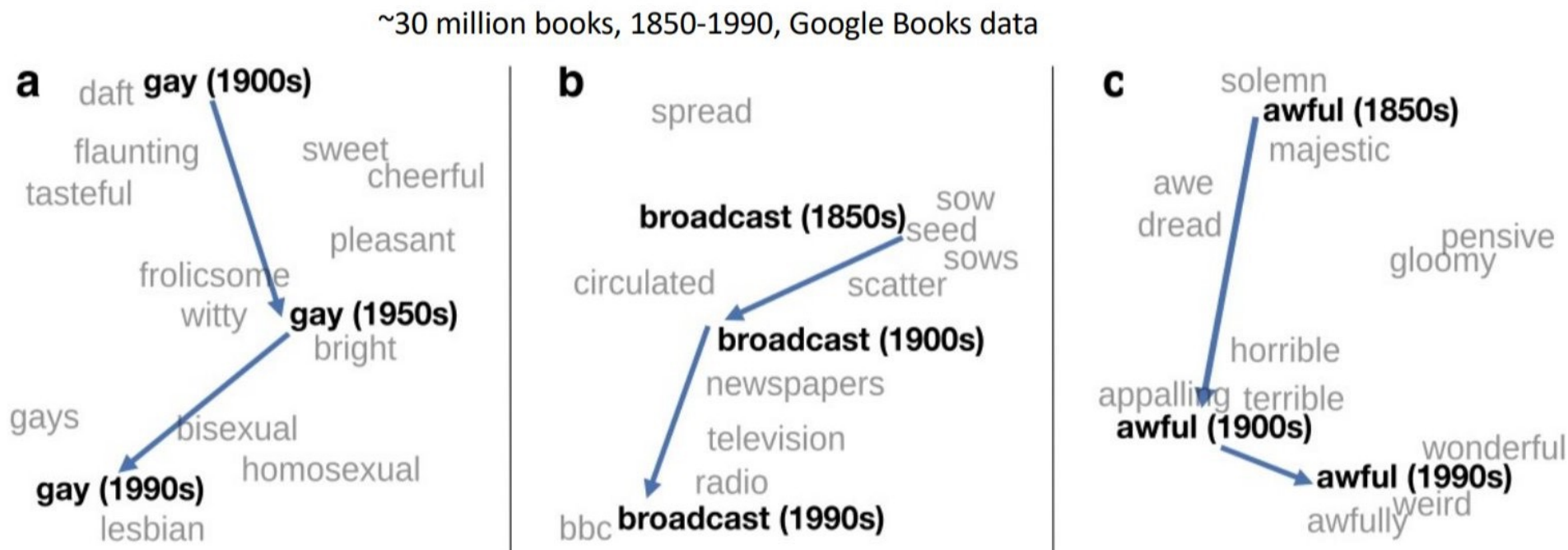
- 问题:
 - “apple is to tree as grape is to _”
 - 解法
 - $\overrightarrow{\text{tree}} - \overrightarrow{\text{apple}} + \overrightarrow{\text{grape}} \rightarrow \text{vine}$



- 其他例子 (Turney and Littman 2005, Mikolov et al. 2013b)
 - $\overrightarrow{\text{king}} - \overrightarrow{\text{man}} + \overrightarrow{\text{woman}} \approx \text{queen}$
 - $\overrightarrow{\text{Paris}} - \overrightarrow{\text{France}} + \overrightarrow{\text{Italy}} \approx \text{Rome}$

词向量可以作为历史语义的窗口

- 用不同年代的历史文本训练词向量/词嵌入，看看意义的变化



William L. Hamilton, Jure Leskovec, and Dan Jurafsky. 2016. Diachronic Word Embeddings Reveal Statistical Laws of Semantic Change. Proceedings of ACL.

词嵌入反映文化偏见！

- 问题: Paris : France :: Tokyo : x
 - x = Japan
- 问题: father : doctor :: mother : x
 - x = nurse
- 问题: man : computer programmer :: woman : x
 - x = homemaker

Bolukbasi, Tolga, Kai-Wei Chang, James Y. Zou, Venkatesh Saligrama, and Adam T. Kalai. "Man is to computer programmer as woman is to homemaker? debiasing word embeddings." In NeurIPS, pp. 4349-4357. 2016

算法中使用词嵌入，如程序员的招聘搜索，可能会导致招聘中的偏见

前馈网络的用例

- 最先进的系统使用更强大的神经架构，如Transformer，但简单的模型也是有用的！

自然语言处理与序列数据

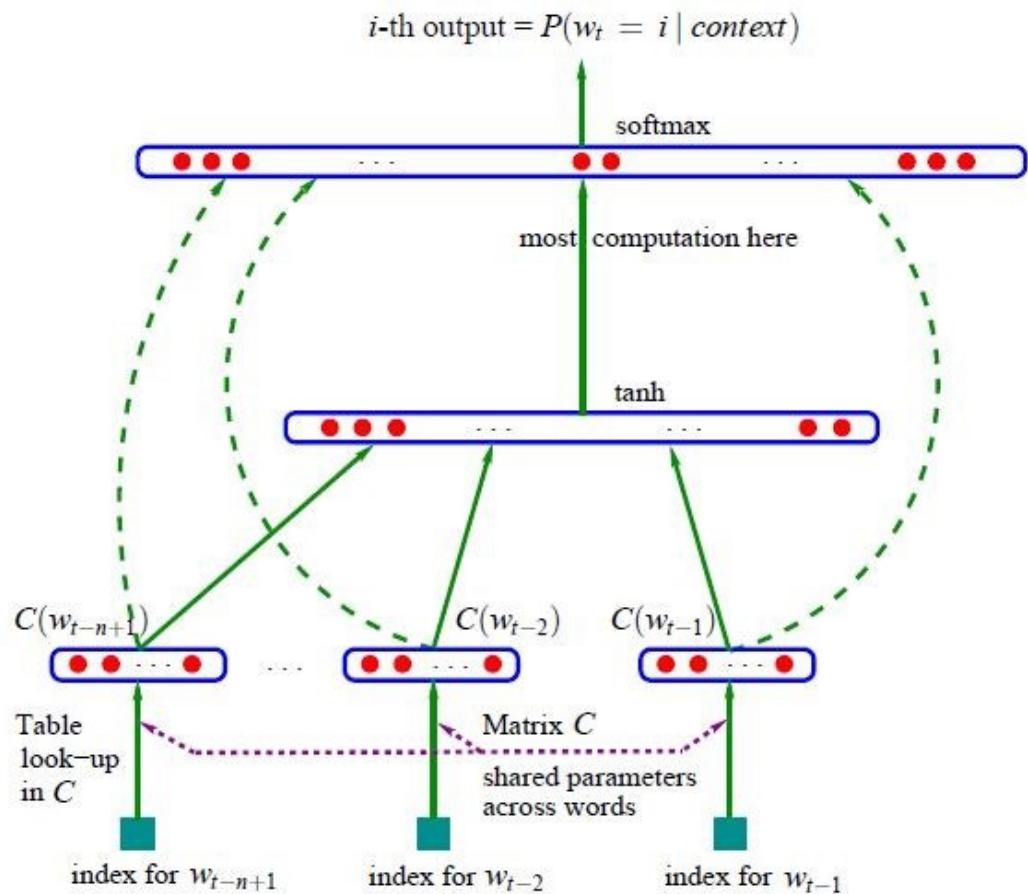
NLP and Sequential Data

- 序列数据在自然语言处理领域中随处可见

- 篇章中的句子 (discourse)
- 句子中的词
- 词中的字符
- ...



神经语言模型



	n	c	h	m	direct	mix	train.	valid.	test.
MLP1	5		50	60	yes	no	182	284	268
MLP2	5		50	60	yes	yes		275	257
MLP3	5		0	60	yes	no	201	327	310
MLP4	5		0	60	yes	yes		286	272
MLP5	5		50	30	yes	no	209	296	279
MLP6	5		50	30	yes	yes		273	259
MLP7	3		50	30	yes	no	210	309	293
MLP8	3		50	30	yes	yes		284	270
MLP9	5		100	30	no	no	175	280	276
MLP10	5		100	30	no	yes		265	252
Kneser-Ney back-off	3							334	323
Kneser-Ney back-off	4							332	321
Kneser-Ney back-off	5							332	321

Yoshua Bengio, Réjean Ducharme, Pascal Vincent, Christian Janvin: A Neural Probabilistic Language Model. J. Mach. Learn. Res. 3: 1137-1155 (2003)

语言的远距离依赖

Long-distance Dependencies in Language

- 数目、性别等的一致性
 - **He** does not have very much confidence in **himself**.
 - **She** does not have very much confidence in **herself**.
- 选择性偏好 Selectional preference
 - The **reign** has lasted as long as the life of the **queen**.
 - The **rain** has lasted as long as the life of the **clouds**.

困难案例

- 指代消解 Anaphora resolution
 - The trophy would not fit in the brown suitcase because **it** was too **big**.
 - The trophy would not fit in the brown suitcase because **it** was too **small**.

(from Winograd Schema Challenge:
<http://commonsensereasoning.org/winograd.html>)

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如果文本输入长度不定，如何处理？

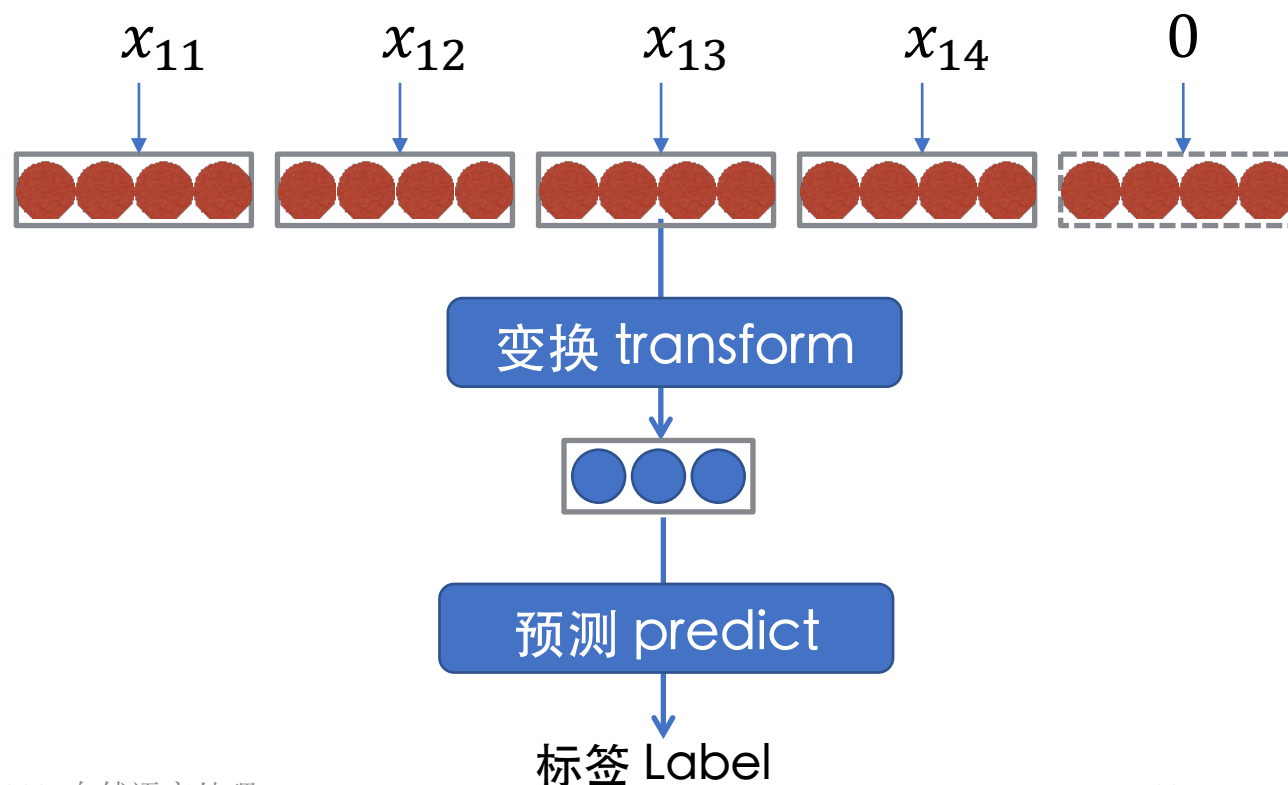
- 输入

- $x_1 = (x_{11}, x_{12}, x_{13}, x_{14})$
- $x_2 = (x_{21}, x_{22}, x_{23})$
- $x_3 = (x_{31}, x_{32}, x_{33}, x_{34}, x_{35})$

+: 简单，紧要关头使用

-: 不适合长文本

- 简单想法：将短序列用零填充到最长序列的长度

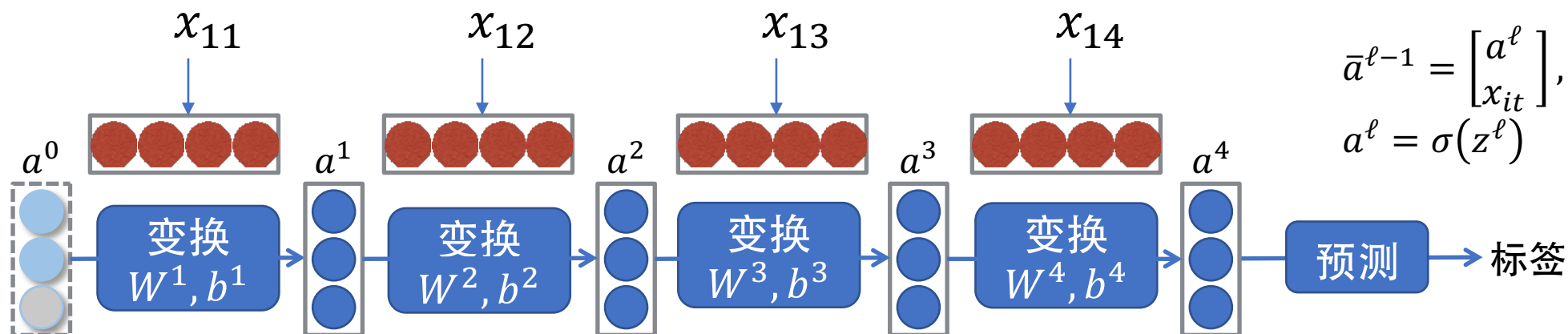


其他方案?

- 输入

- $x_1 = (x_{11}, x_{12}, x_{13}, x_{14})$
- $x_2 = (x_{21}, x_{22}, x_{23})$
- $x_3 = (x_{31}, x_{32}, x_{33}, x_{34}, x_{35})$

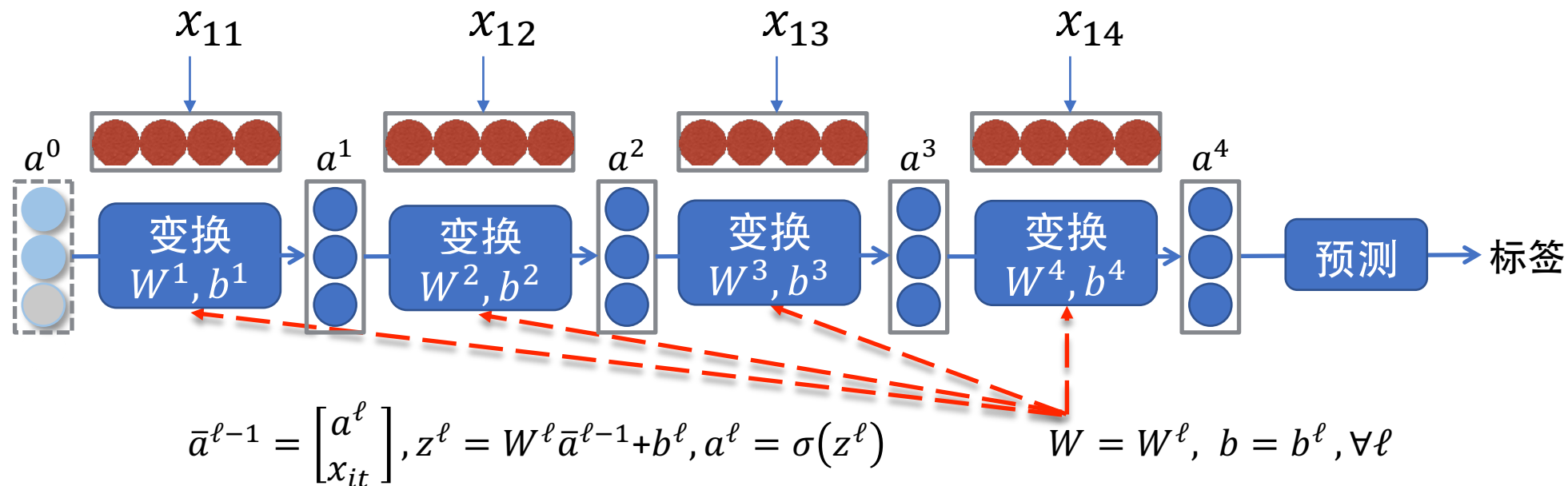
- 新思路: 向每个前馈层输入一个单词



- + : 不需要填充句子
- + : 每个前馈层比连接输入的巨型前馈层小得多
- + : 句子越短需要的层数越少
- : 显而易见的问题: 长句问题

$$\bar{a}^{\ell-1} = \begin{bmatrix} a^\ell \\ x_{it} \end{bmatrix}, z^\ell = W^\ell \bar{a}^{\ell-1} + b^\ell$$
$$a^\ell = \sigma(z^\ell)$$

共享参数？



+: 我们想要多少层就有多少层

• 这称为循环神经网络 (Recurrent Neural Network, RNN) – “可变深度” 网络

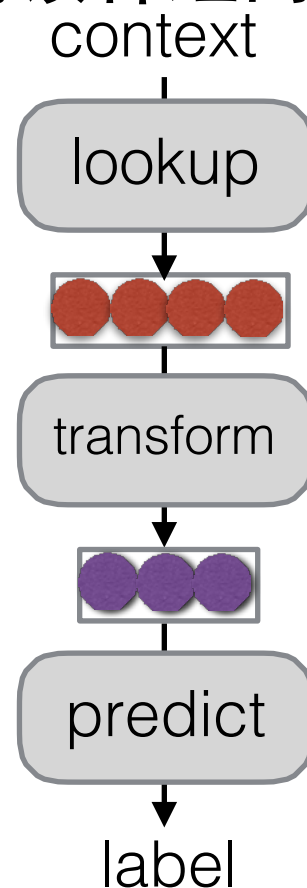
循环神经网络

Recurrent Neural Networks

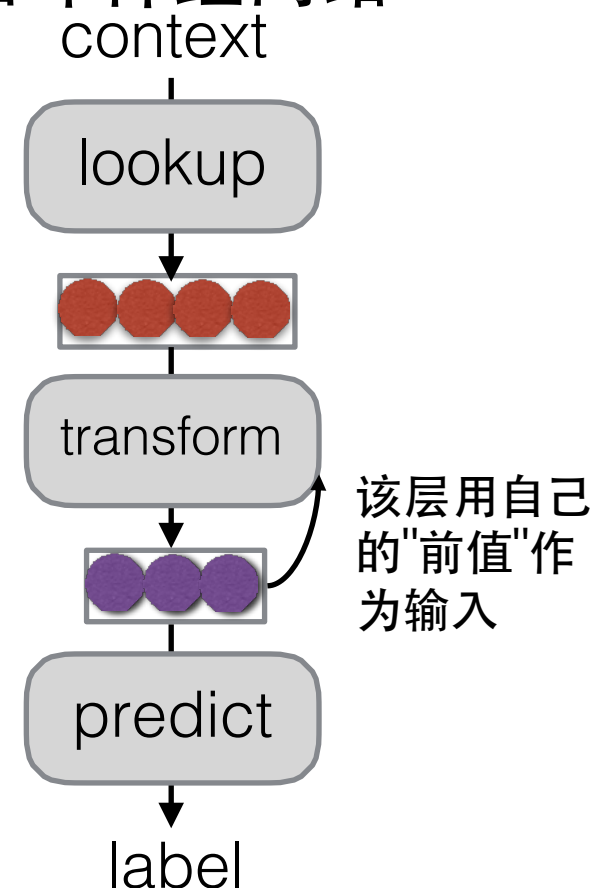
(Elman 1990)

- 沿着“**时间维度**”“**记忆**”信息的工具
- 循环神经网络就是跨层共享权值的神经网络，每层均有一个输入，且层数可变

前馈神经网络

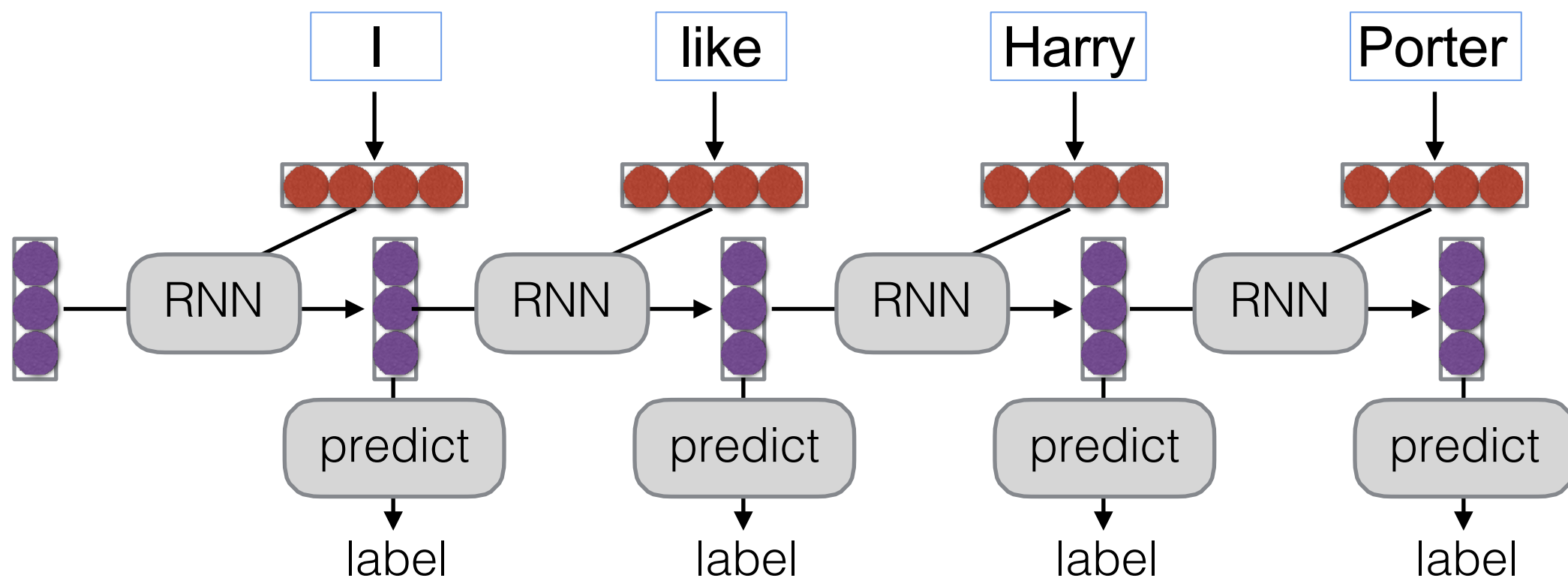
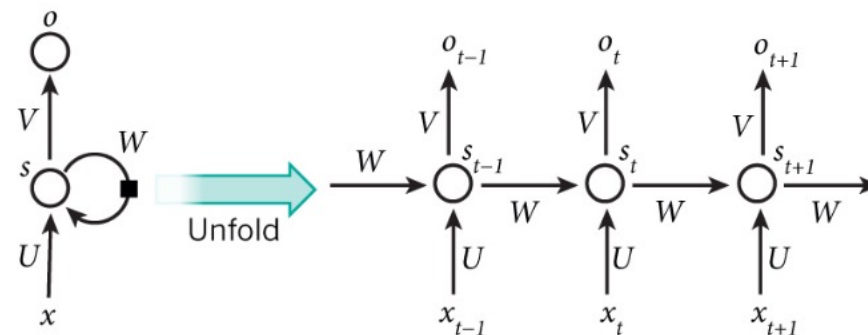


循环神经网络

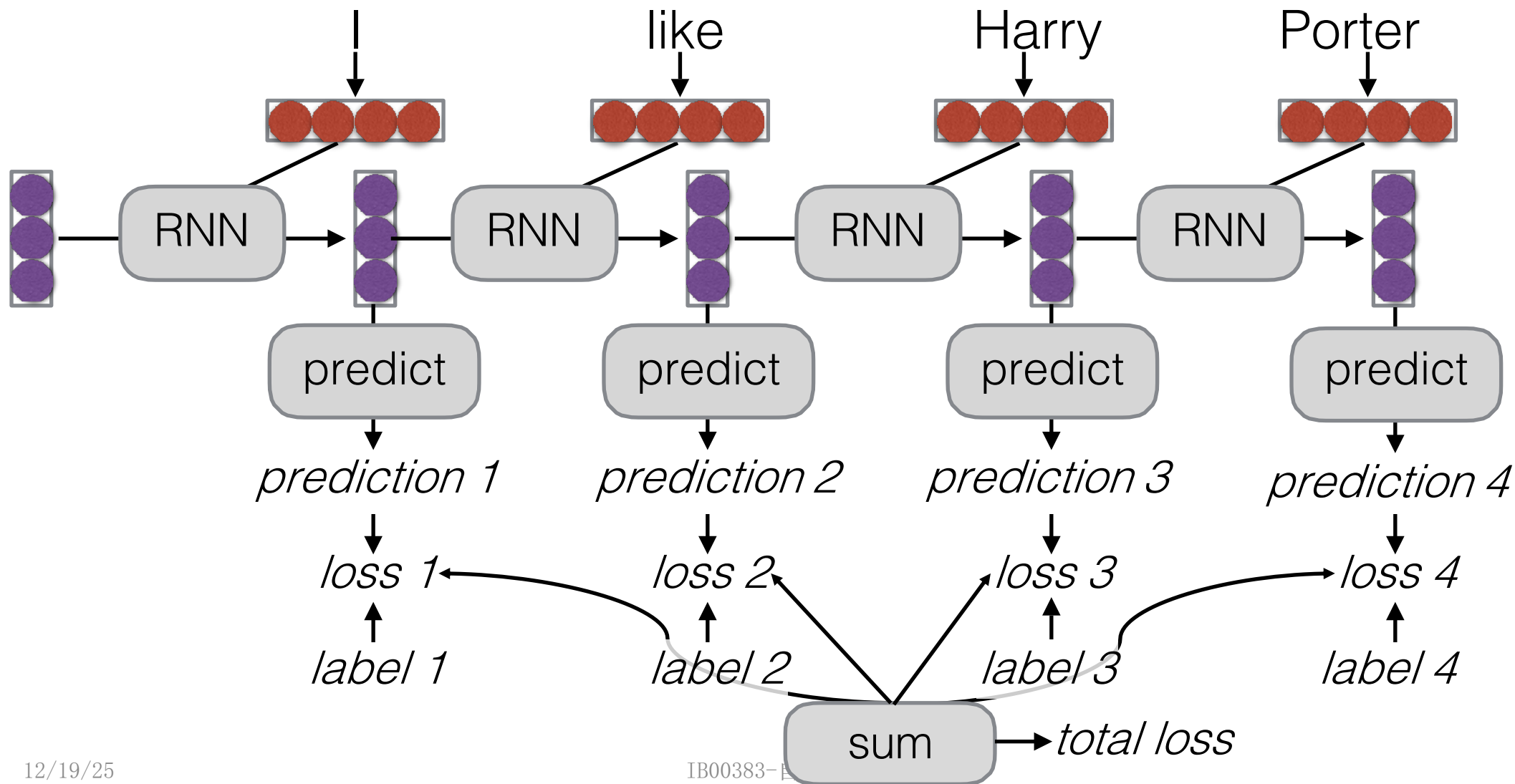


沿时间展开 Unrolling in Time

- 处理序列的过程

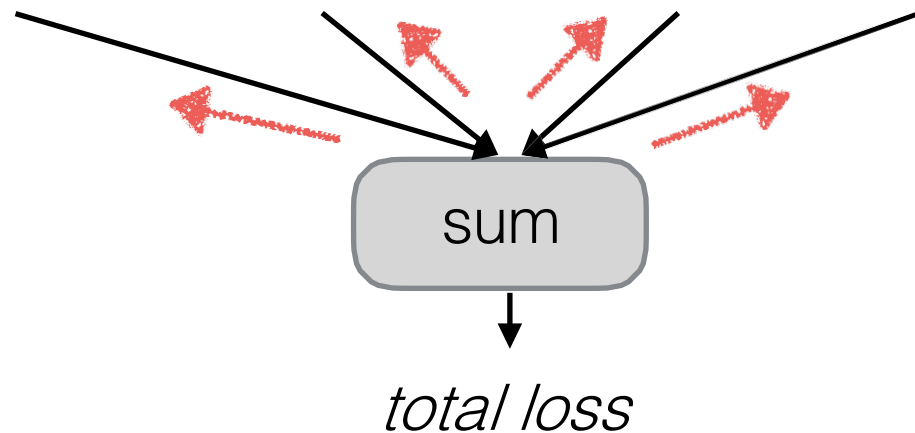


循环神经网络的训练



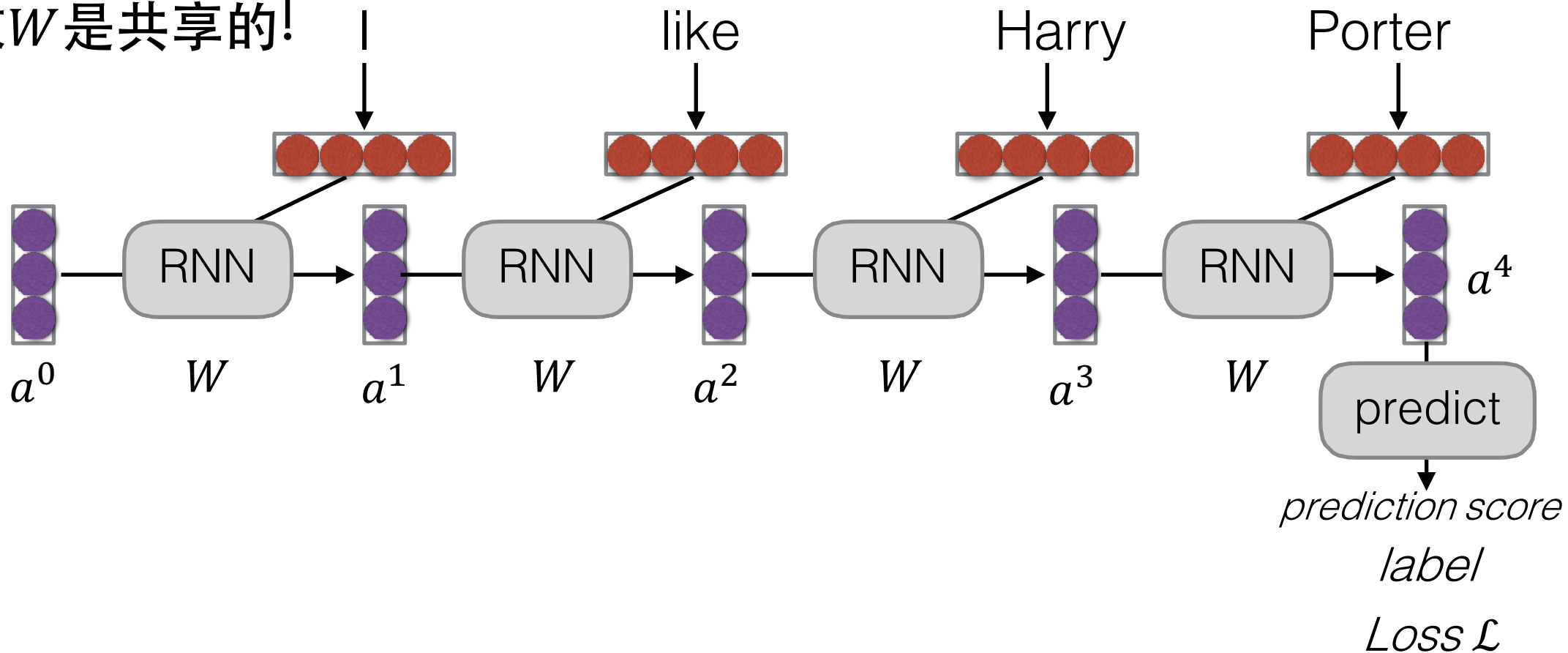
循环神经网络的训练

- 展开图是结构良好的有向无环图(Directed Acyclic Graph, DAG) -- 我们可以采用反向传播更新参数
- 参数在不同时间步共享，导数(梯度)由所有时间步上汇总所得
- 该过程历称为“时间反向传播”(BackPropagation Through Time, BPTT)

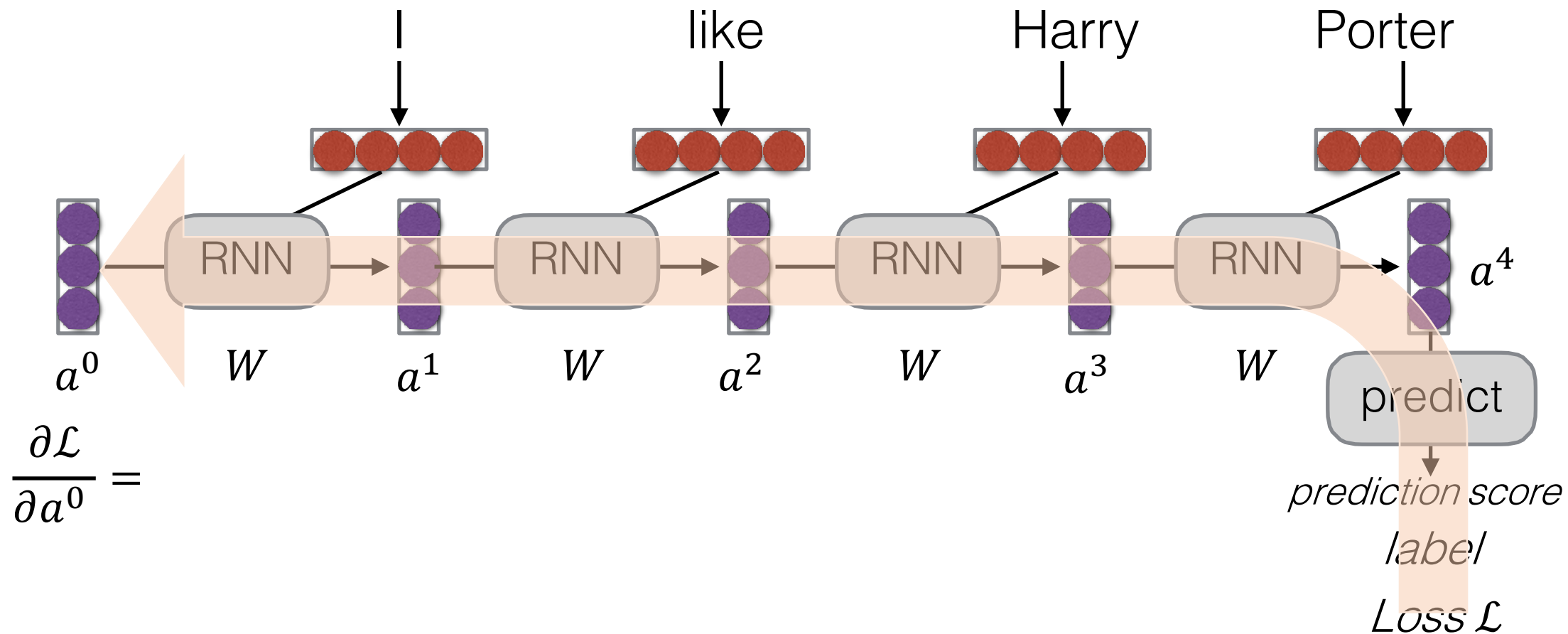


前向传播

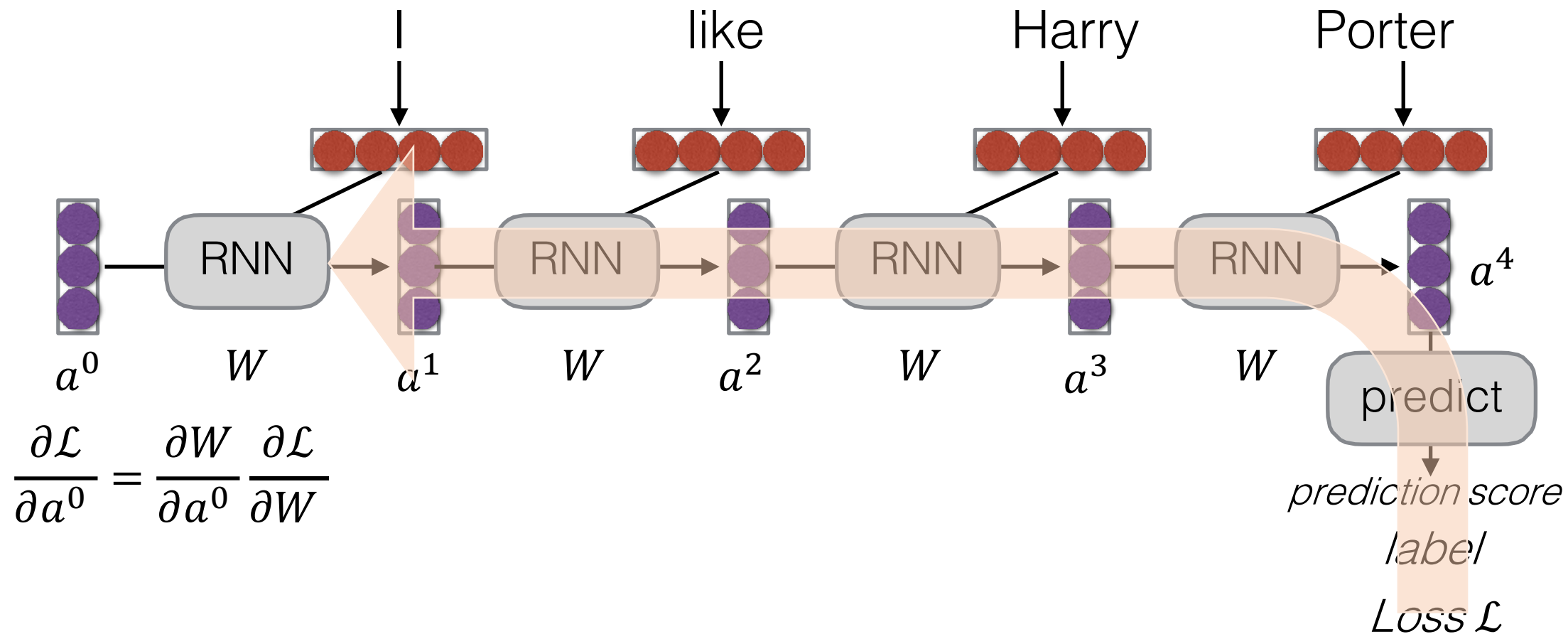
- 参数 W 是共享的!



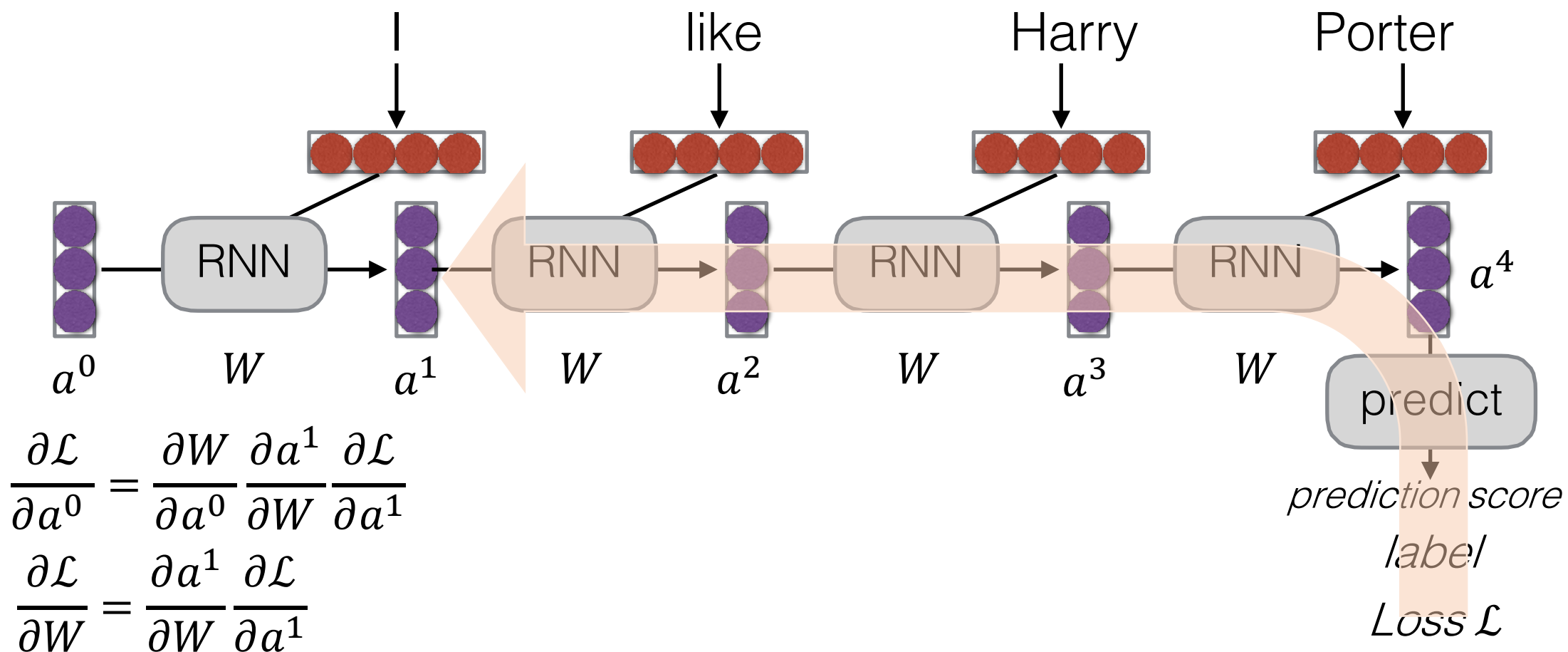
反向传播



反向传播

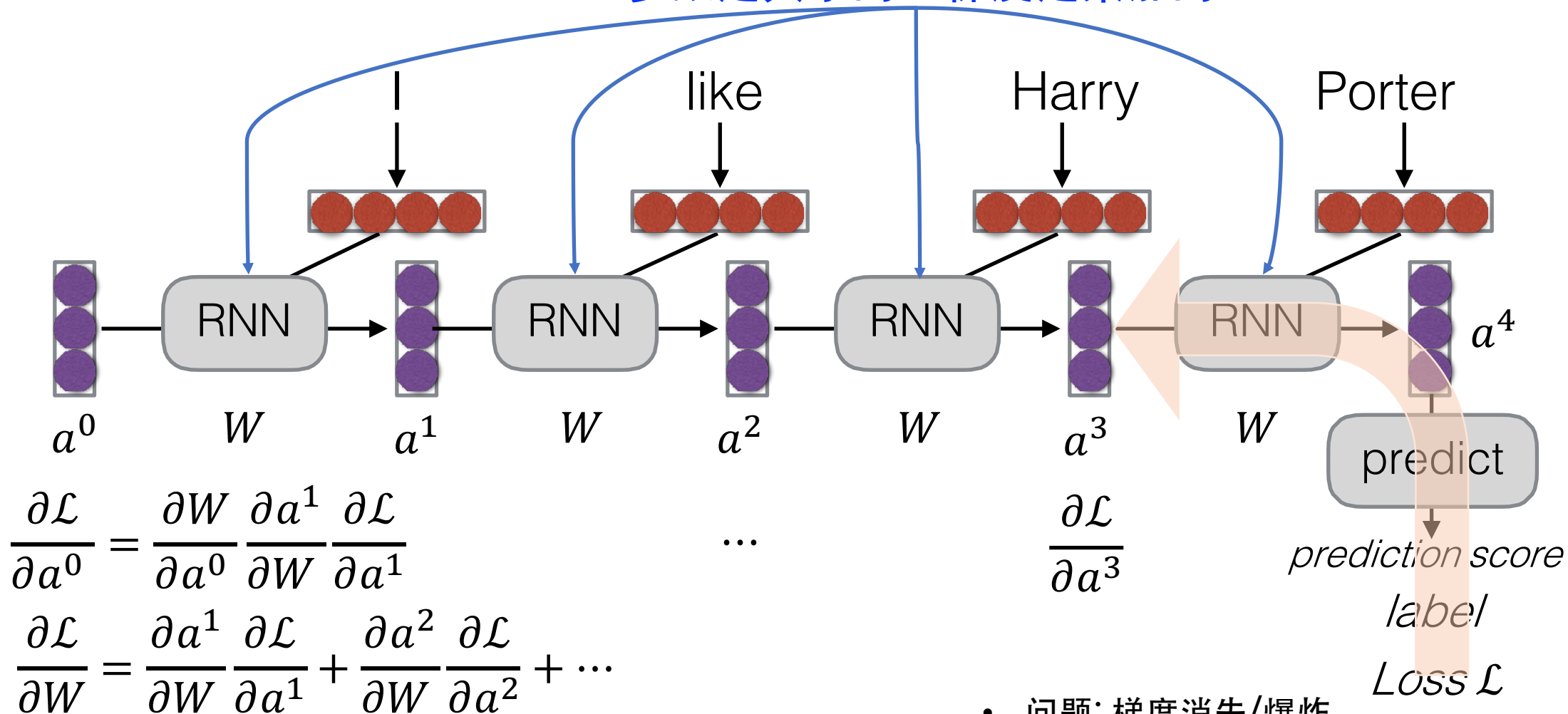


反向传播



反向传播

参数是共享的！梯度是累加的！



• 问题: 梯度消失/爆炸

大纲

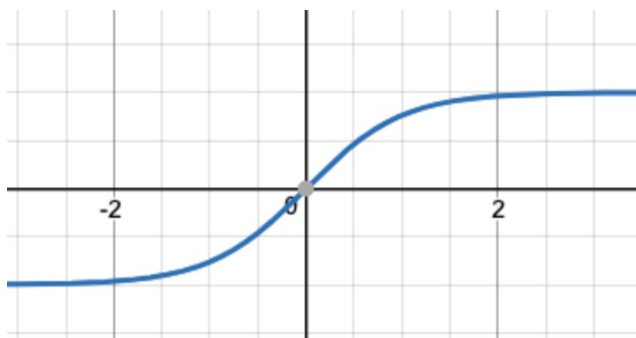
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为什么梯度消失是个问题？

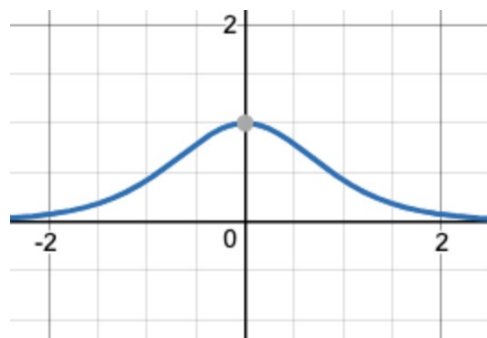
Gradient Vanishing

- Why?

- 由非线性激活函数(如双曲正切函数)的**压缩效应/squashed**，或是权重矩阵中小**权重的连乘**所导致的



$$f(x) = \tanh(x)$$



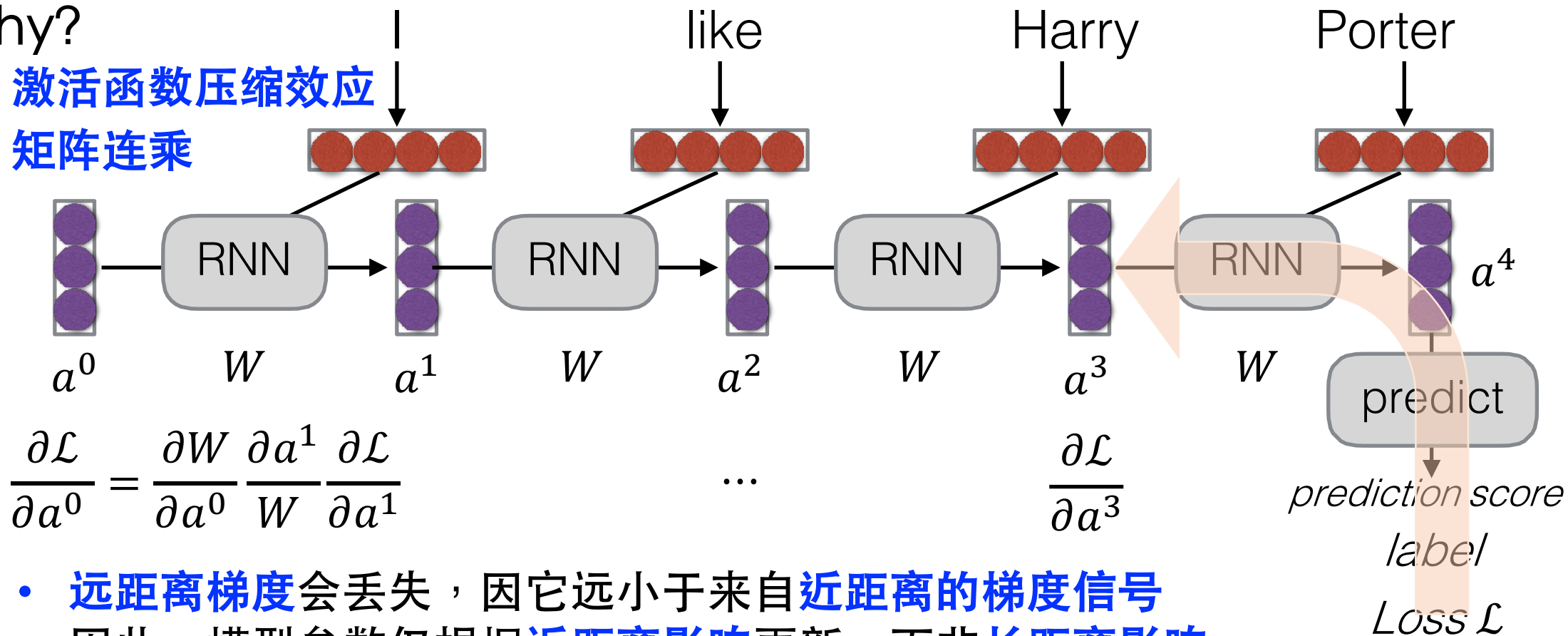
$$\frac{\partial f}{\partial x} = 1 - \tanh(x)^2$$

$$\frac{\partial \mathcal{L}}{\partial a^0} = \frac{\partial W}{\partial a^0} \frac{\partial a^1}{\partial W} \frac{\partial \mathcal{L}}{\partial a^1} \dots$$

为什么梯度消失是个问题？ Gradient Vanishing

- Why?

- 激活函数压缩效应
- 矩阵连乘



- 远距离梯度会丢失，因它远小于来自近距离的梯度信号
- 因此，模型参数仅根据近距离影响更新，而非长距离影响

方案：长短期记忆网络

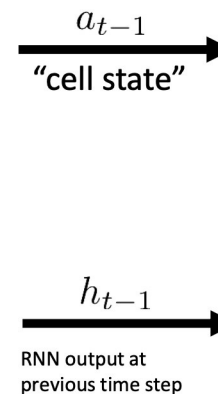
(Hochreiter and Schmidhuber 1997)

Long Short-term Memory (LSTM)

- **核心思想**：在时间步之间建立**加法连接**
- 加法操作不会改变上一时间步的梯度，避免梯度消失的问题
- 通过**门控机制**控制信息流
- 采用两种类型的记忆单元，以不同的节奏对历史上下文信息进行汇总

长短期记忆网络

- 实际情况，LSTM比原始循环神经网络效果好
 - 长期记忆单元: $a_{t-1} \iff$ 短期记忆状态: h_{t-1}



长短期记忆网络

- 实际情况，LSTM比原始循环神经网络效果好
 - 长期记忆单元: $a_{t-1} \Leftrightarrow$ 短期记忆状态: h_{t-1}
 - 遗忘门 f_t : 决定需要遗忘多少历史记忆信息
 - 输入门 i_t : 决定向记忆单元中添加多少新信息
 - g_t : 计算需要添加到记忆单元中的新值
 - 输出门 o_t : 决定有多少信息被映射到下一时刻的状态中

$$f_t = \sigma(\bar{f}_t)$$

$$i_t = \sigma(\bar{i}_t)$$

$$g_t = \tanh(\bar{g}_t)$$

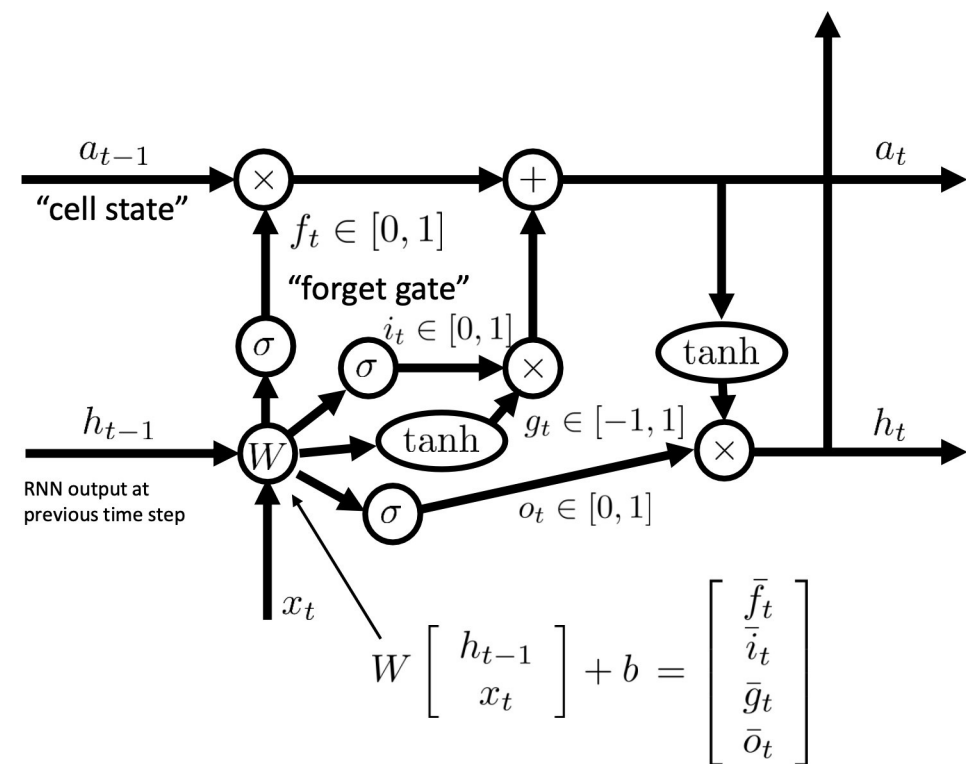
$$o_t = \sigma(\bar{o}_t)$$

σ : sigmoid

$$f_t, i_t, g_t, o_t \in \mathbb{R}^d$$

$$\text{LSTM: } W \in \mathbb{R}^{4d \times 2d}$$

$$\text{原始循环神经网络: } W \in \mathbb{R}^{d \times 2d}$$



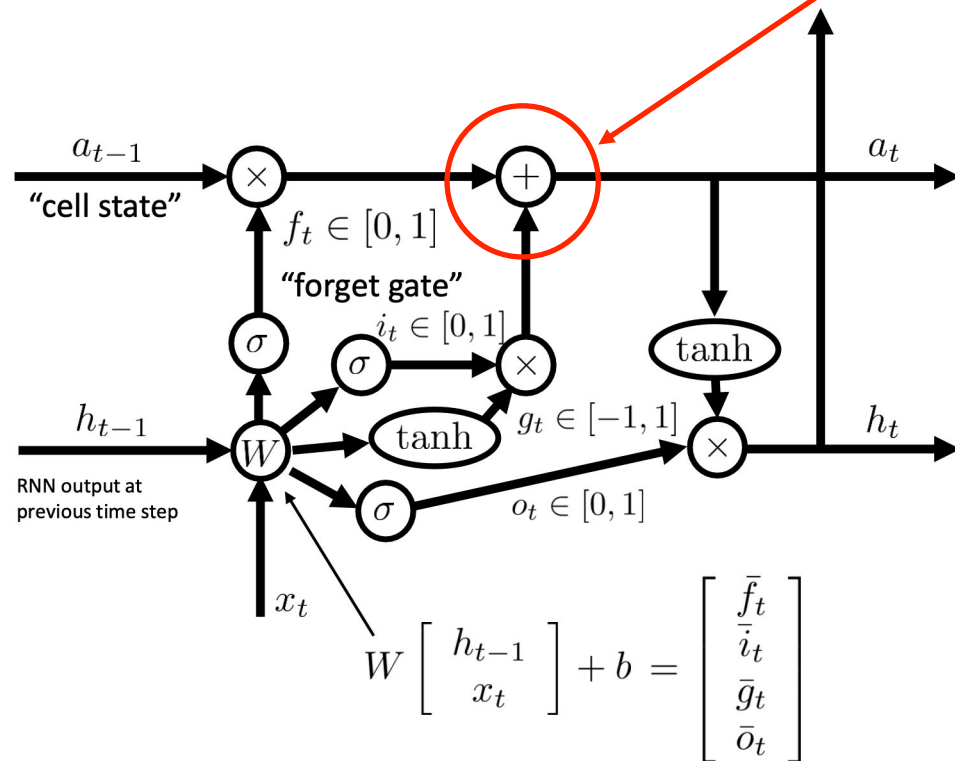
$$a_t = f_t \otimes a_{t-1} + i_t \otimes g_t$$

$$h_t = o_t \otimes \tanh(a_t)$$

$\otimes$ 表示元素乘积

长短期记忆网络

这个加法操作是点睛之笔！



$$a_t = f_t \otimes a_{t-1} + i_t \otimes g_t$$

随时间步缓慢更新
“长期记忆”

$$h_t = o_t \otimes \tanh(a_t)$$

一直在更新(乘法运算)
“短期记忆”

为什么梯度爆炸是个问题？

Gradient Exploding

- 如果梯度值变得过大，随机梯度下降(SGD)的参数更新步长也会随之过大：

$$\theta \leftarrow \theta - \eta \frac{\partial \mathcal{L}}{\partial \theta}$$

- 这会导致**糟糕的参数更新**
 - 如果步长过大，模型会得到混乱且糟糕的参数
- 最坏情况下，这会引发网络中的**Inf**或**NaN**错误
 - 你不得不回退到之前的checkpoint，重新开始训练

方案：梯度裁剪 Gradient Clipping

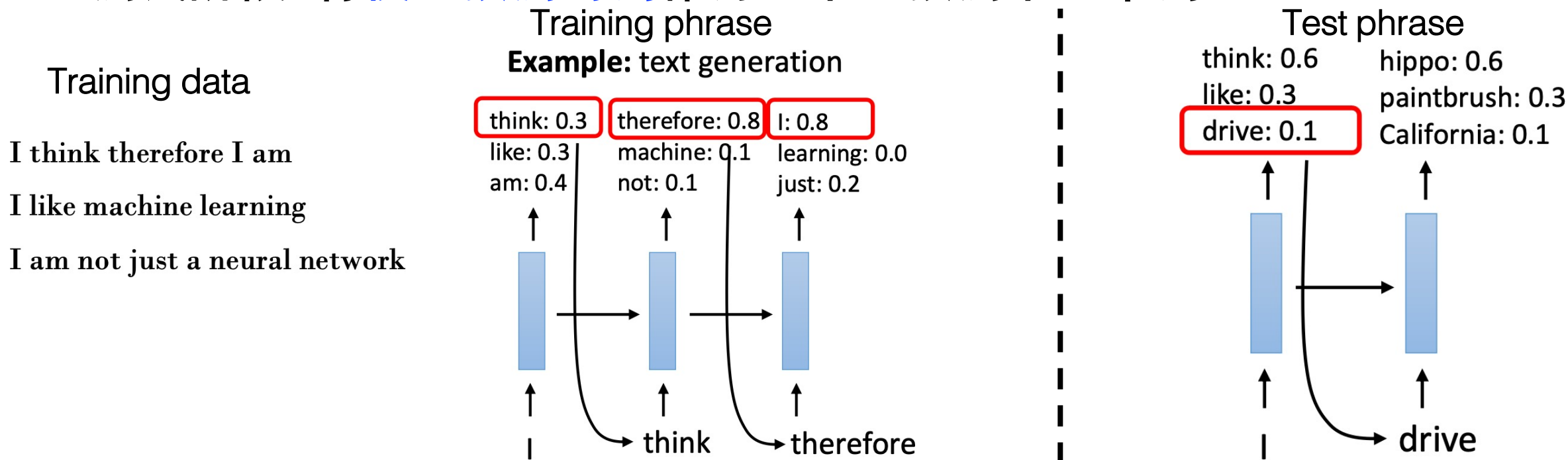
- 若梯度的范数大于某个阈值，则在执行随机梯度下降(SGD)更新前，将其按比例缩小
- **设计思路**：沿相同方向更新参数，但采用更小的步长

Algorithm 1 Pseudo-code for norm clipping

```
 $\hat{\mathbf{g}} \leftarrow \frac{\partial \mathcal{E}}{\partial \theta}$   
if  $\|\hat{\mathbf{g}}\| \geq threshold$  then  
     $\hat{\mathbf{g}} \leftarrow \frac{threshold}{\|\hat{\mathbf{g}}\|} \hat{\mathbf{g}}$   
end if
```

补充说明：分布偏移 (或暴露偏差)

- 训练阶段: 将**真实标签词**作为上下文预测下一个词
- 测试阶段: 将**模型预测的词**作为上下文预测下一个词

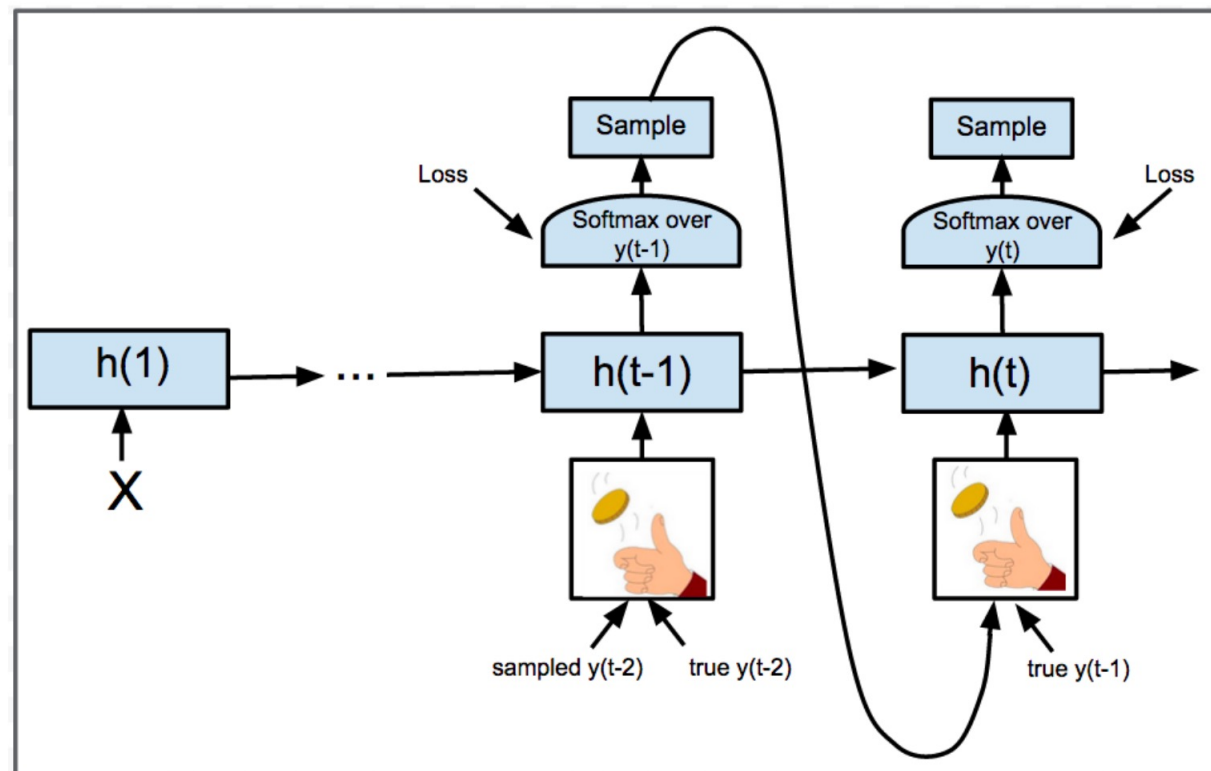
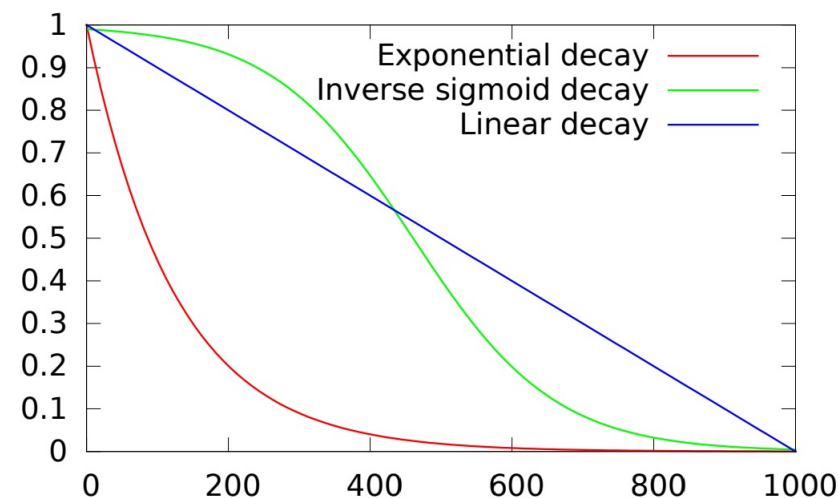


分布偏移: 输入数据的分布从训练时的真实文本序列，偏移到了测试时的预测文本序列
哪怕是一个随机的预测错误，都可能导致输出结果彻底错乱！

补充说明：时序采样

- 借鉴自强化学习，用于循环神经网络训练的经典技巧
 - **训练初期**：输入为真实标签序列
 - 因为此时模型的预测结果大多没有实际意义
 - **训练中期**：随机决定输入真实标签序列还是模型自身的预测结果
 - **训练末期**：主要输入是模型的预测结果，以此缓解分布偏移问题

Samy Bengio, Oriol Vinyals, Navdeep Jaitly, Noam Shazeer:
Scheduled Sampling for Sequence Prediction with Recurrent
Neural Networks. NIPS 2015: 1171-1179



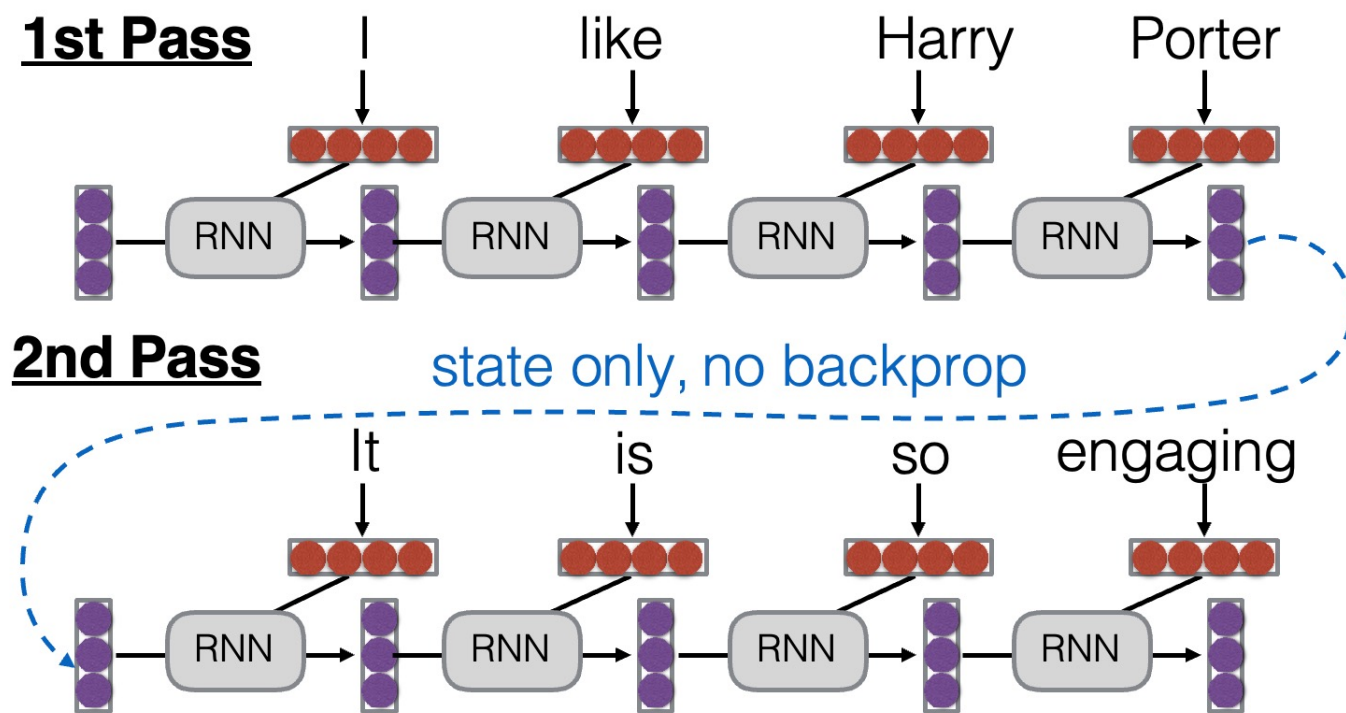
处理长序列文本

- 有时我们希望捕捉长序列中的长期依赖关系，
 - 如，完整文档中的单词序列
- 这类长序列往往无法一次性读入GPU(显存不足)

截断式时序反向传播

Truncated Backpropagation Through Time

- 在较短的序列片段上执行反向传播，并利用前一个片段的状态来初始化当前片段的状态

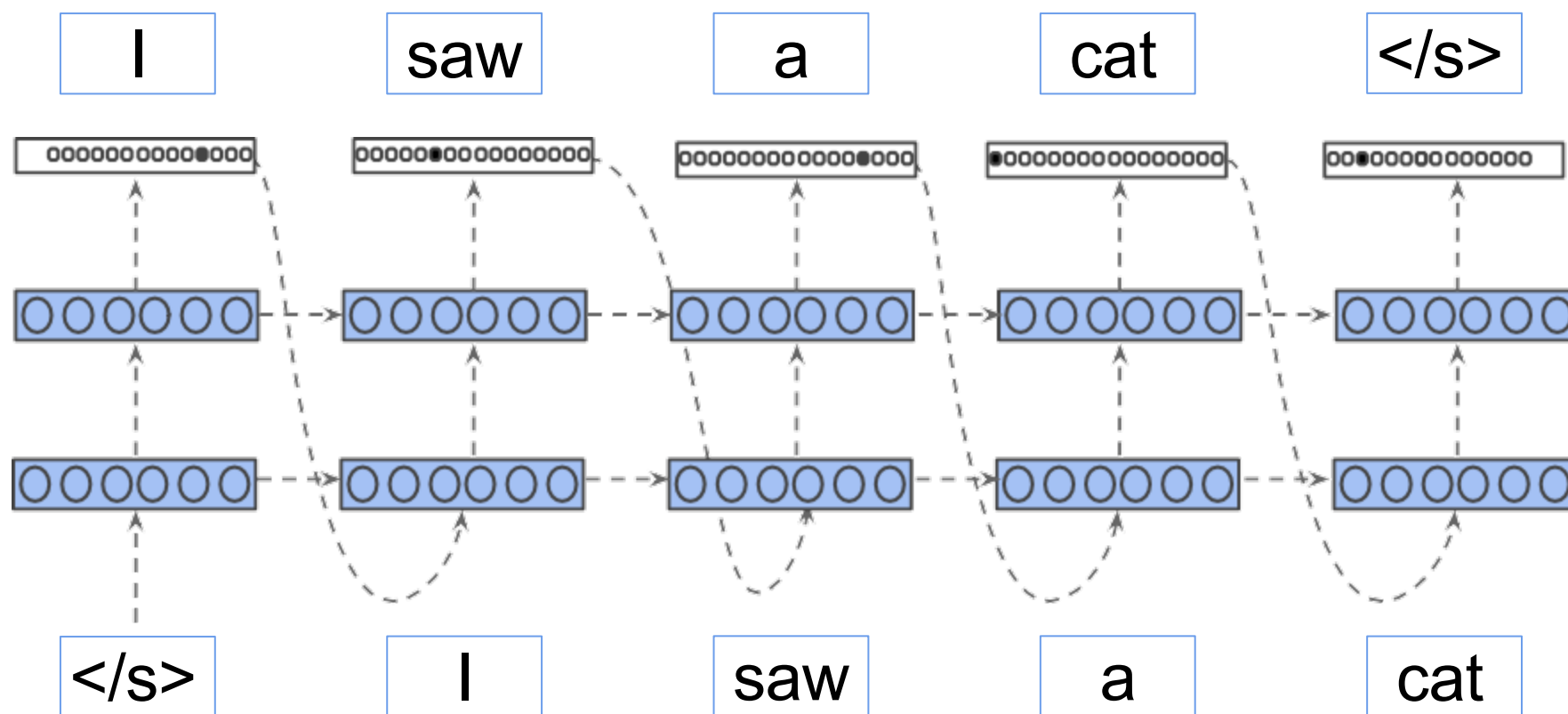


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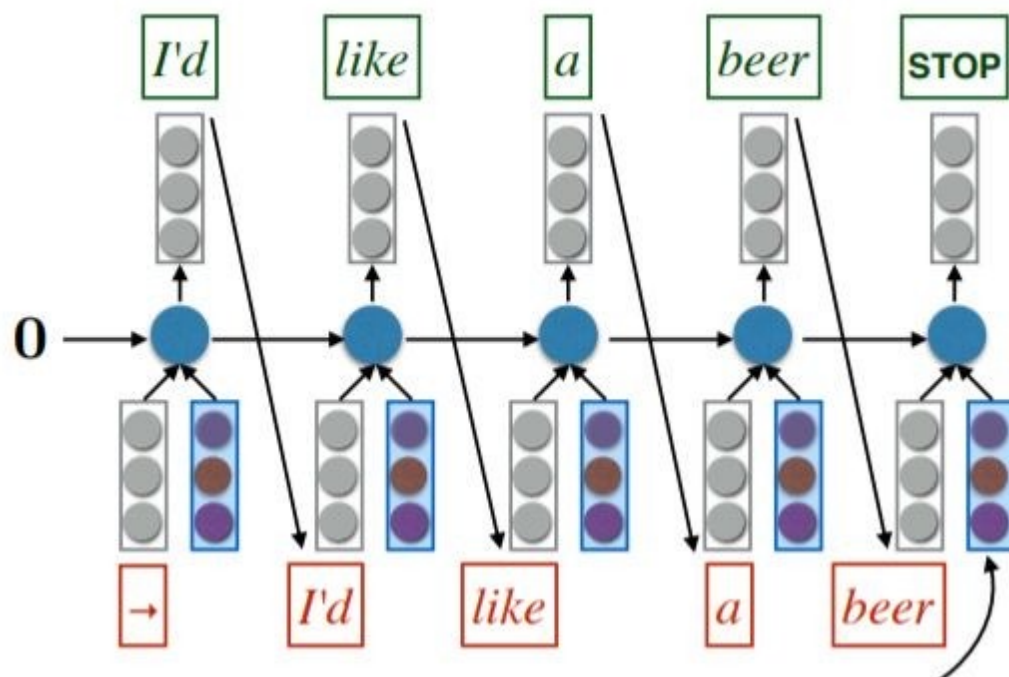
循环语言模型

Recurrent Language Models



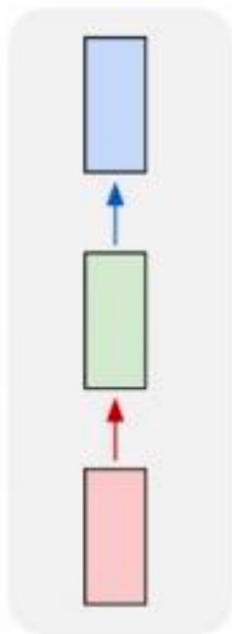
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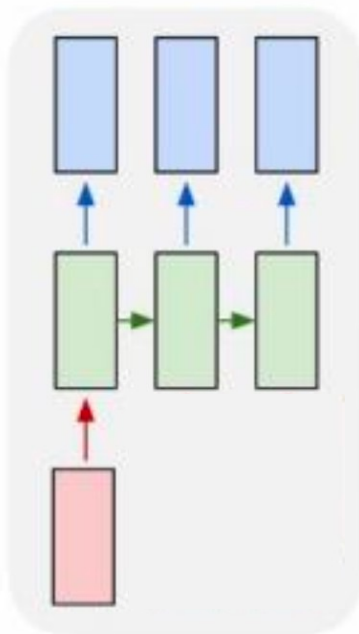
循环神经网络的不同使用方法

one to one



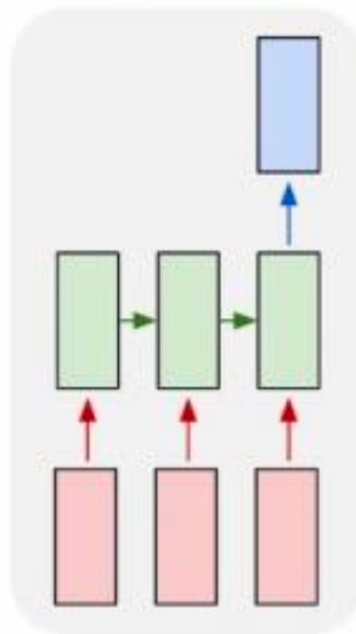
单样本分类
/预测

one to many



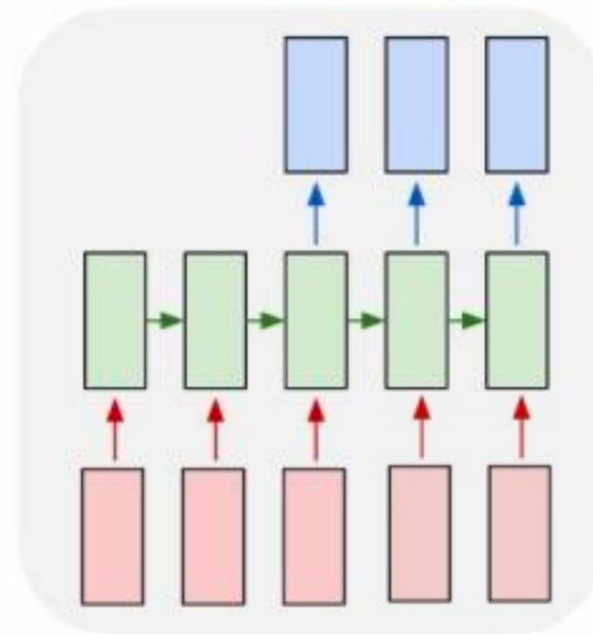
看图说话

many to one



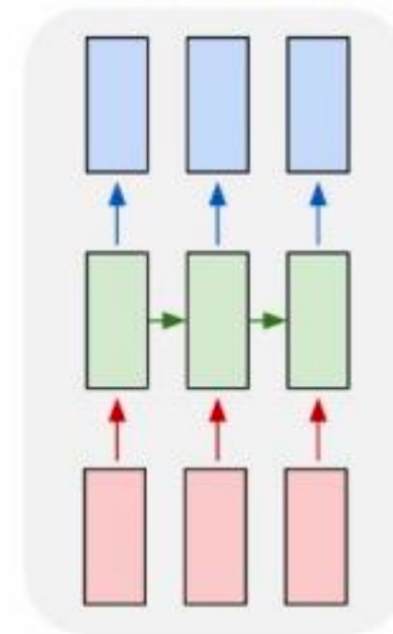
文本分类

many to many



机器翻译

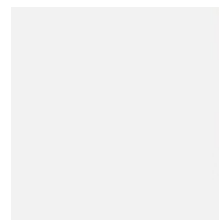
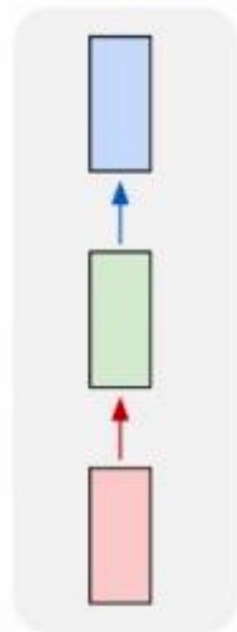
many to many



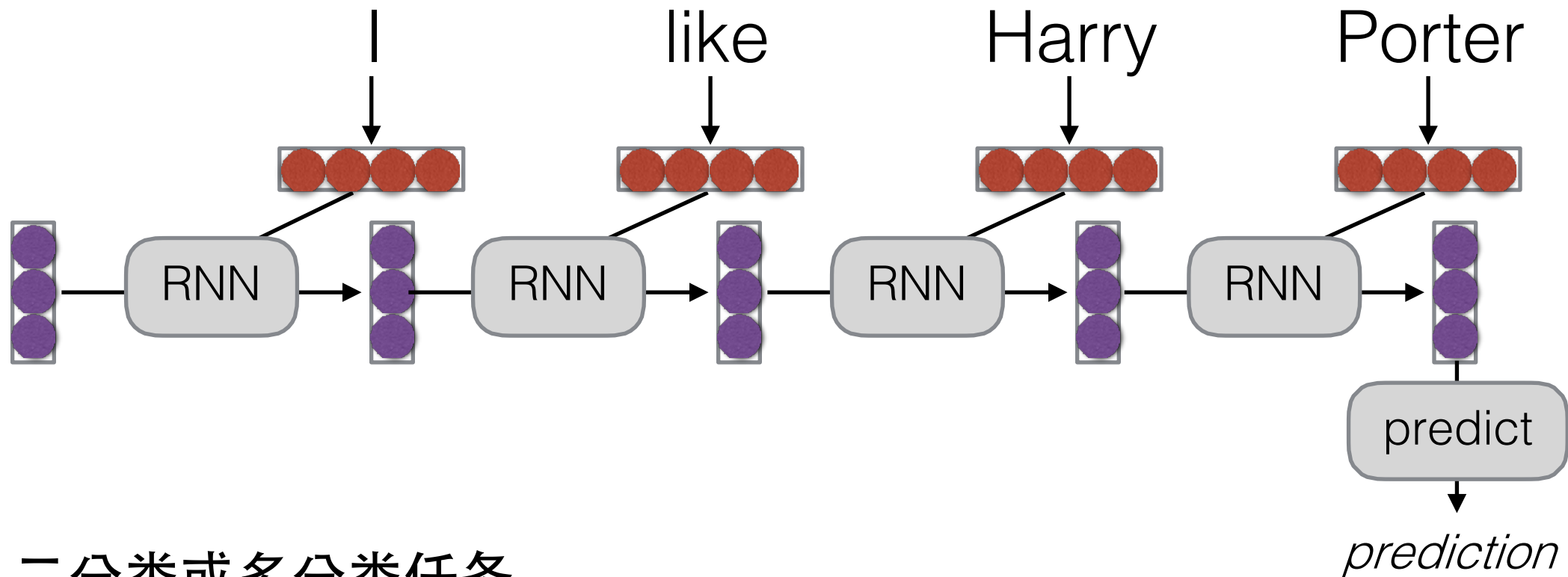
句子标注
(POS, NER)

单样本预测(一对一)

one to one



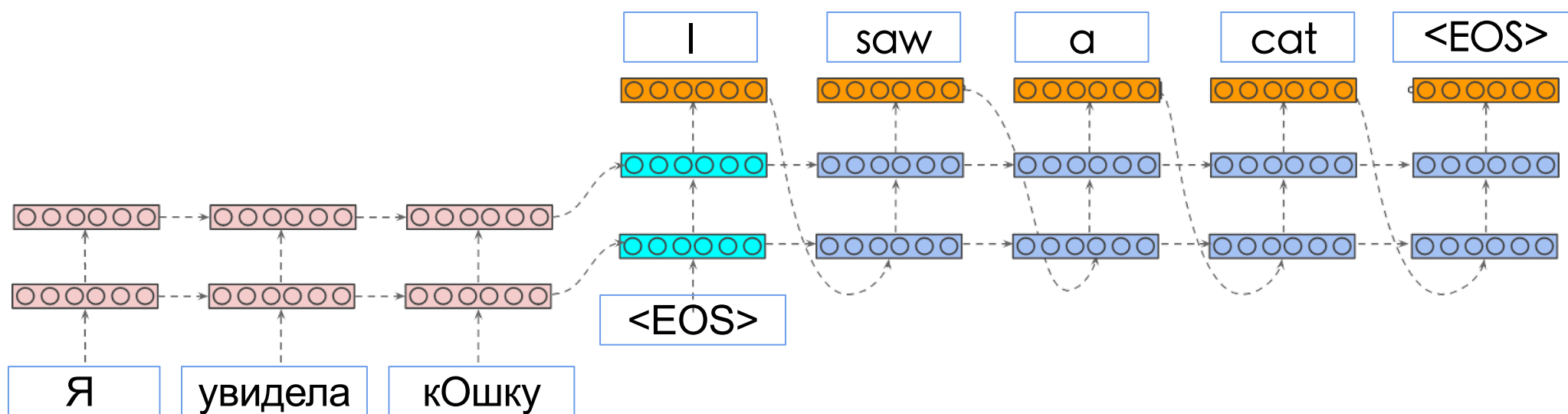
句子编码（多对一）



- 二分类或多分类任务
- 用于检索、句子比对等任务的句子表征

序列到序列模型(多对多，长度不同)

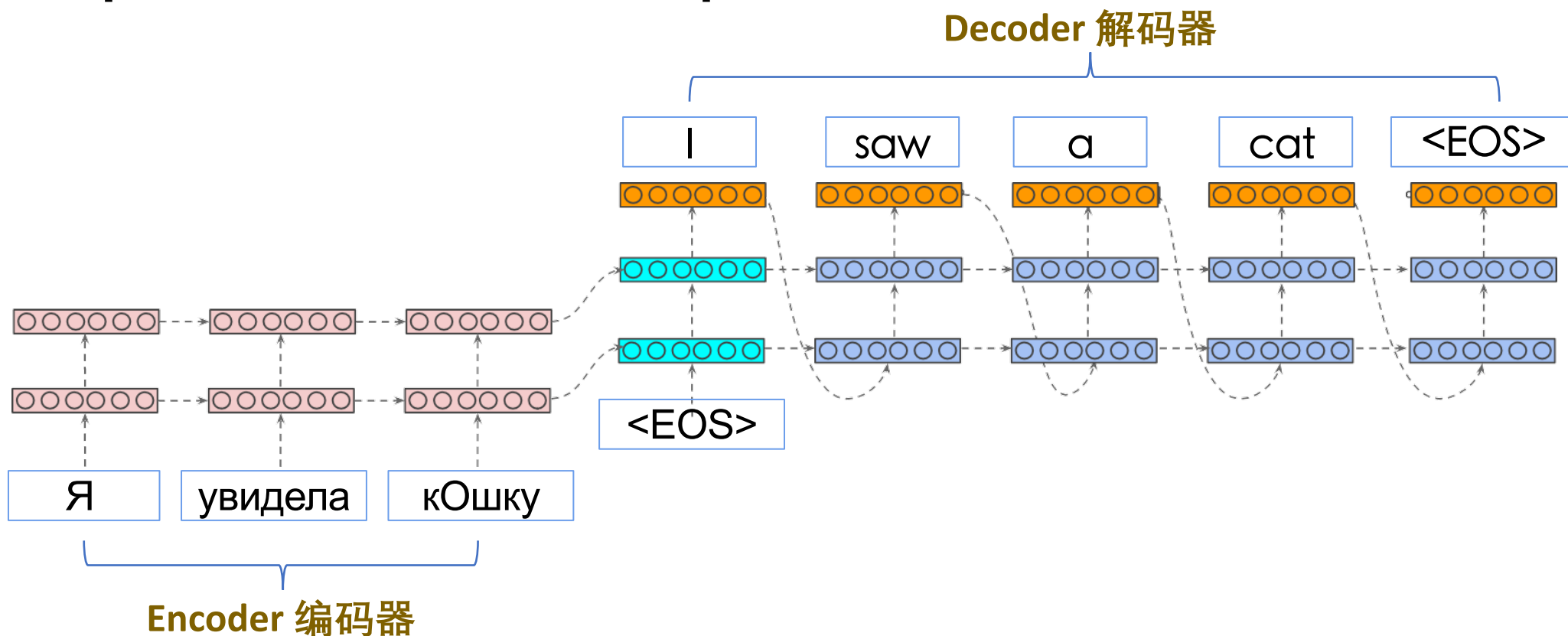
Sequence-to-Sequence Models



Ilya Sutskever, Oriol Vinyals, Quoc V. Le. 2014. Sequence to Sequence Learning with Neural Networks. Proc. NIPS

序列到序列模型(多对多，长度不同)

Sequence-to-Sequence Models

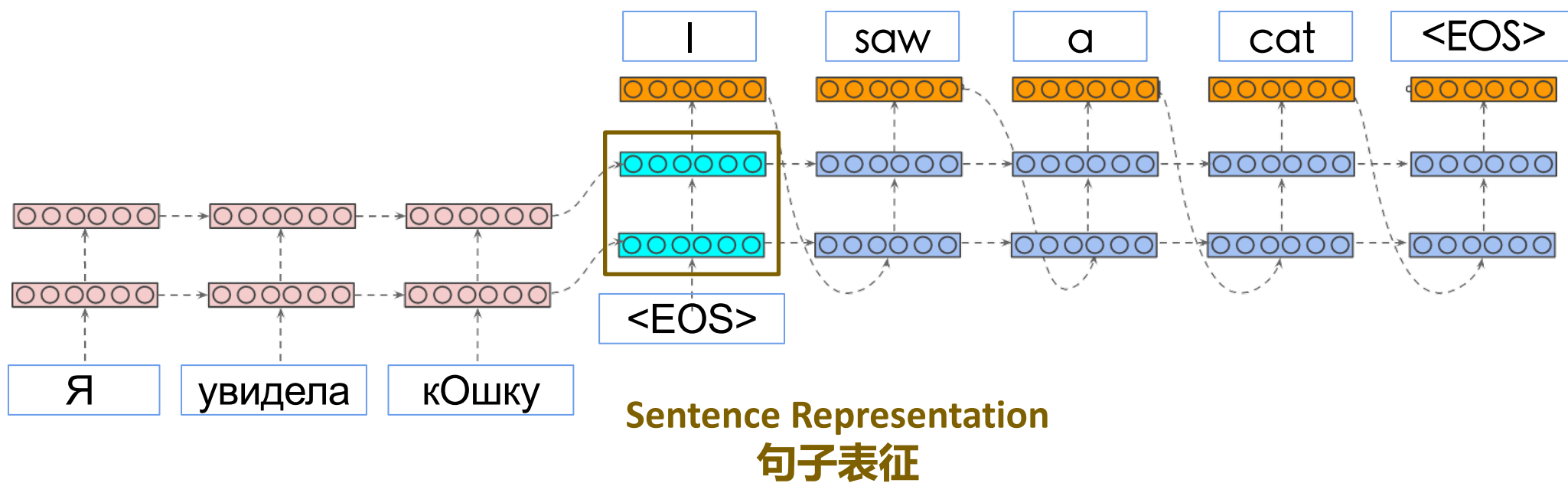


Ilya Sutskever, Oriol Vinyals, Quoc V. Le. 2014. Sequence to Sequence Learning with Neural Networks. Proc. NIPS

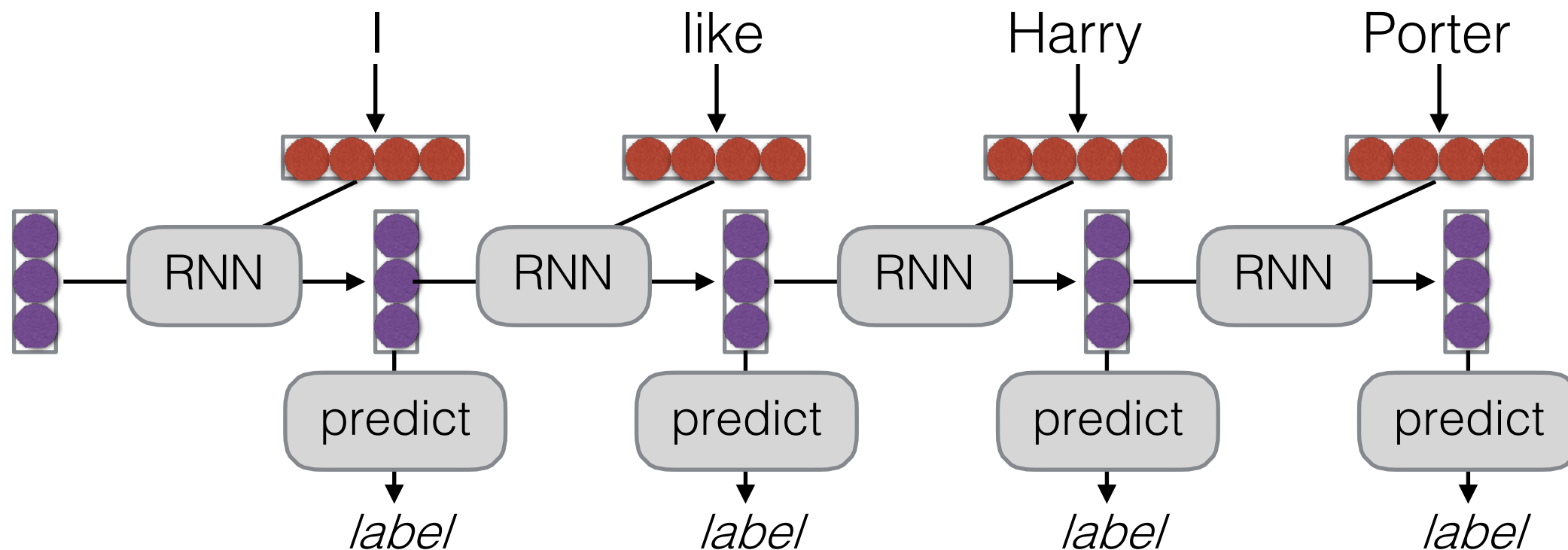
Kyunghyun Cho, Bart van Merriënboer, Dzmitry Bahdanau, Yoshua Bengio. 2014. On the Properties of Neural Machine Translation: Encoder-Decoder Approaches. Proc. SSST

序列到序列模型(多对多，长度不同)

Sequence-to-Sequence Models



序列到序列模型(多对多，长度相同)

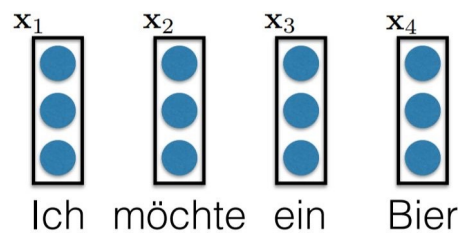


• 序列标注

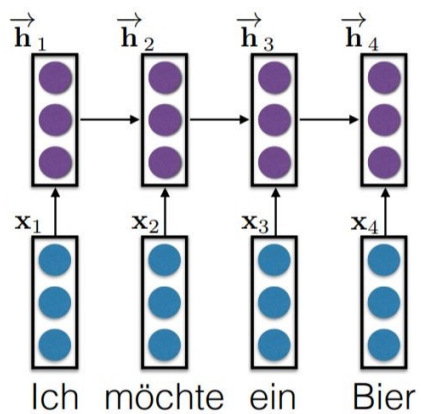
- 词性标注 Part-of-Speech, POS tagging
- 命名实体识别 Named Entity Recognition (NER)

任务	I	like	Harry	Porter
POS	PRP	VBP	NNP	NNP
NER	O	O	B-PER	I-PER

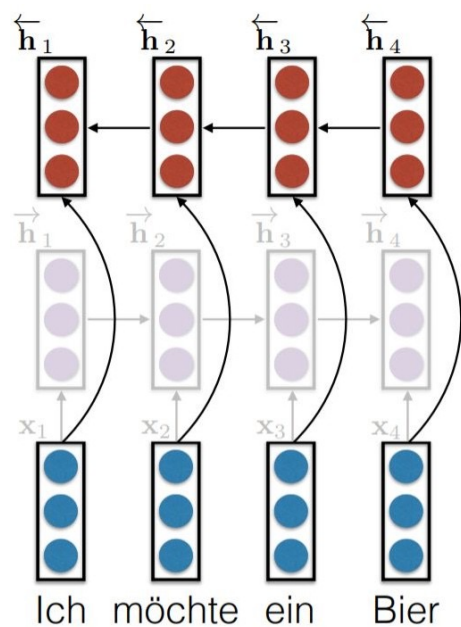
编码器：双向循环神经网络



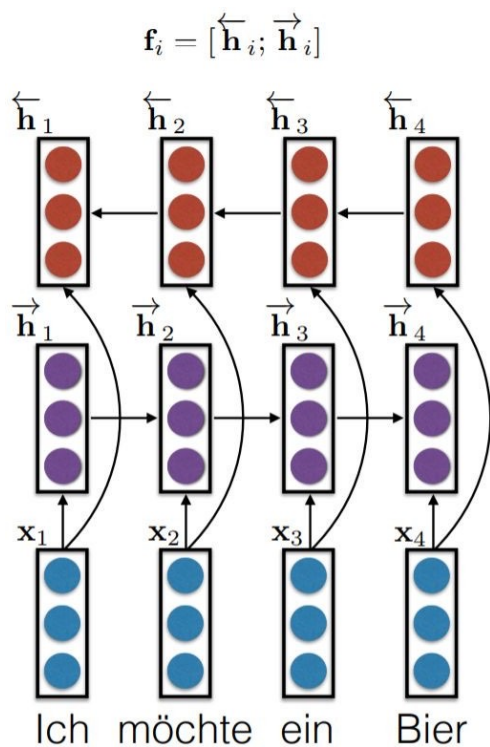
编码器：双向循环神经网络



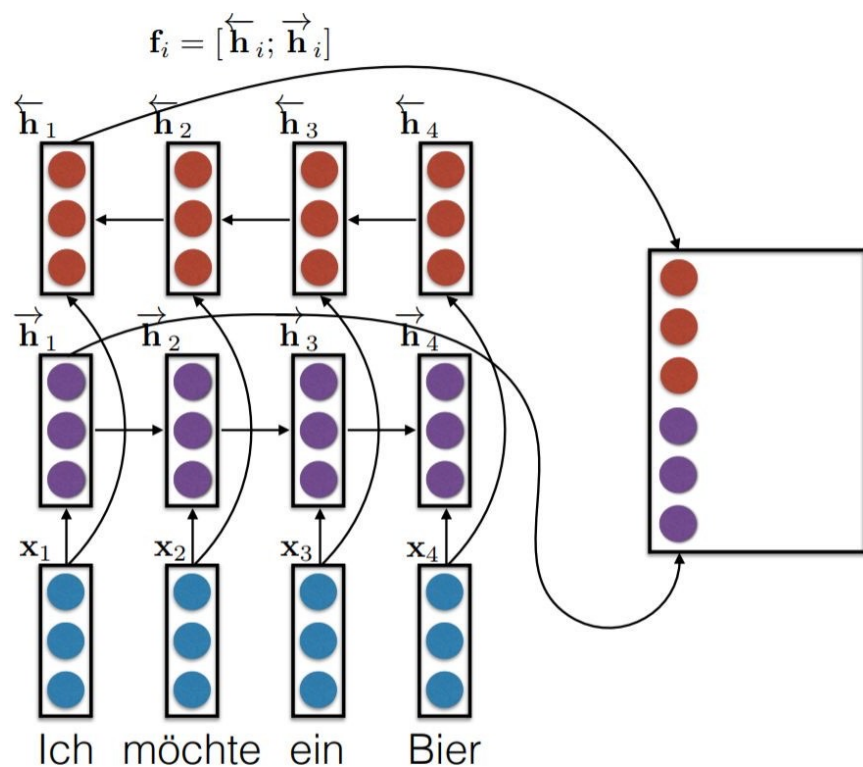
编码器：双向循环神经网络



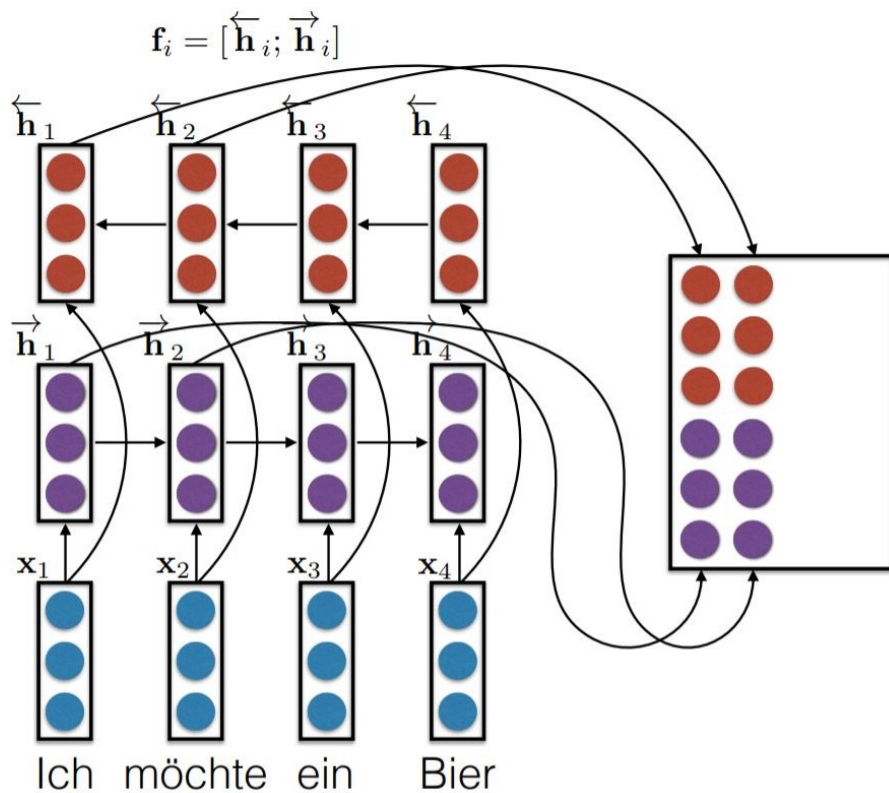
编码器：双向循环神经网络



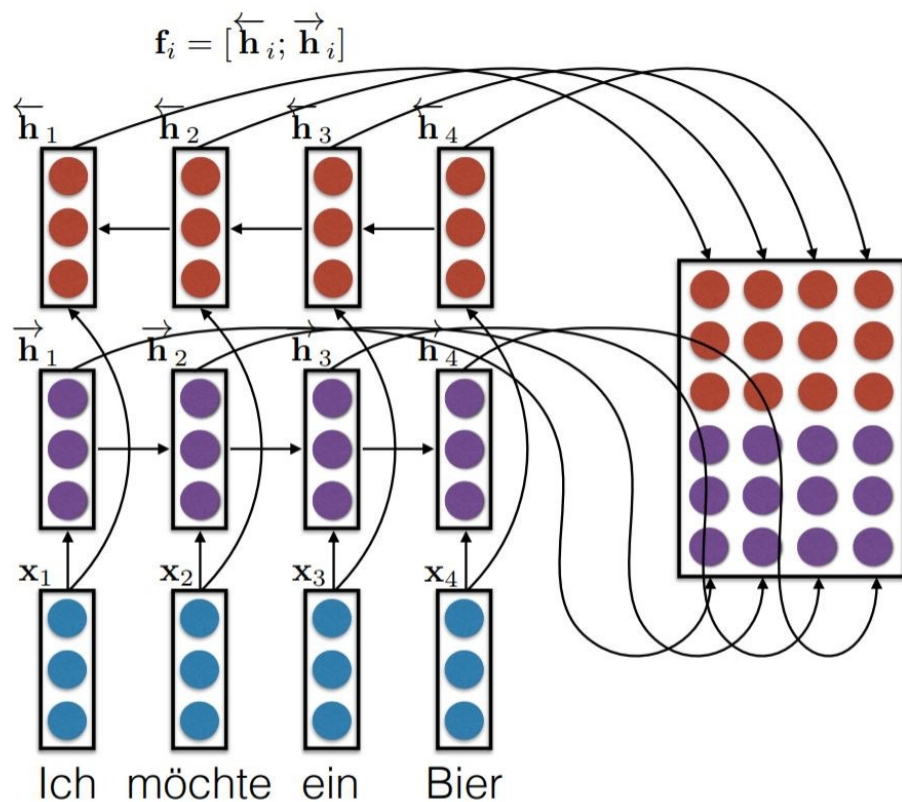
编码器：双向循环神经网络



编码器：双向循环神经网络

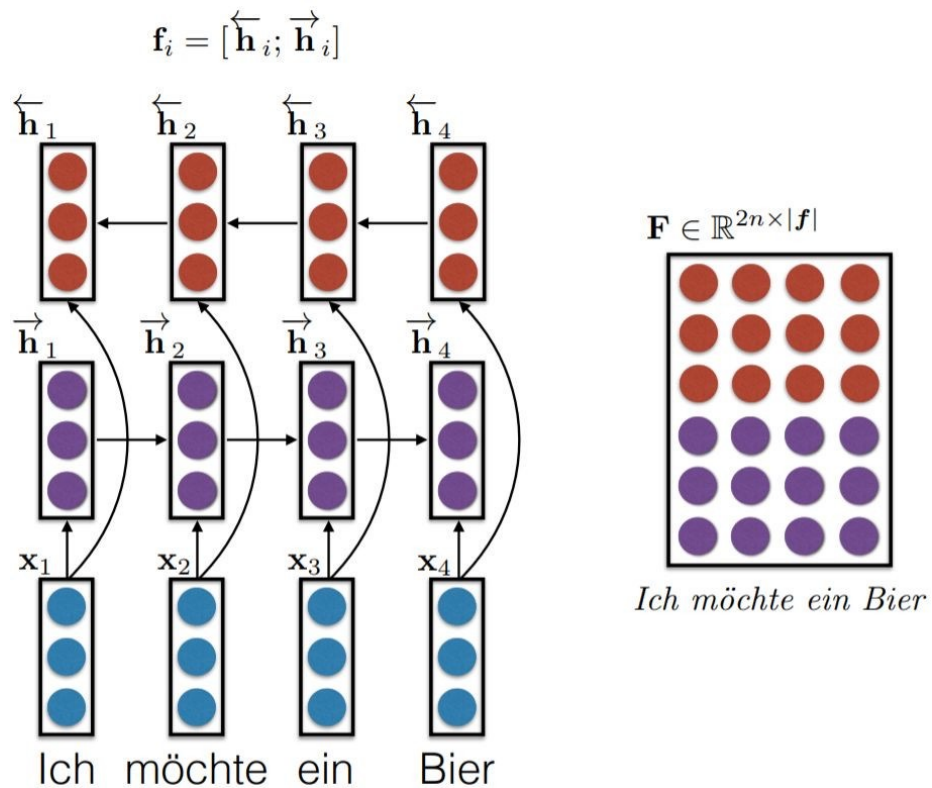


编码器：双向循环神经网络

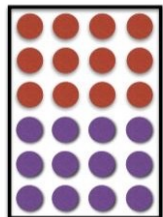
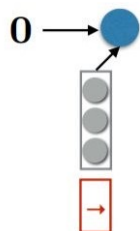


编码器：双向循环神经网络

- 矩阵编码句子 matrix-encoded sentence

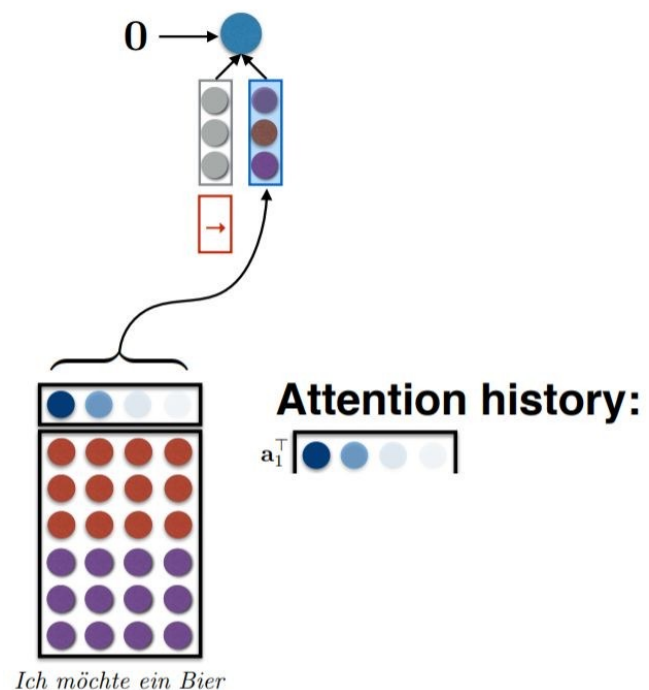


解码器：循环神经网络+注意力机制

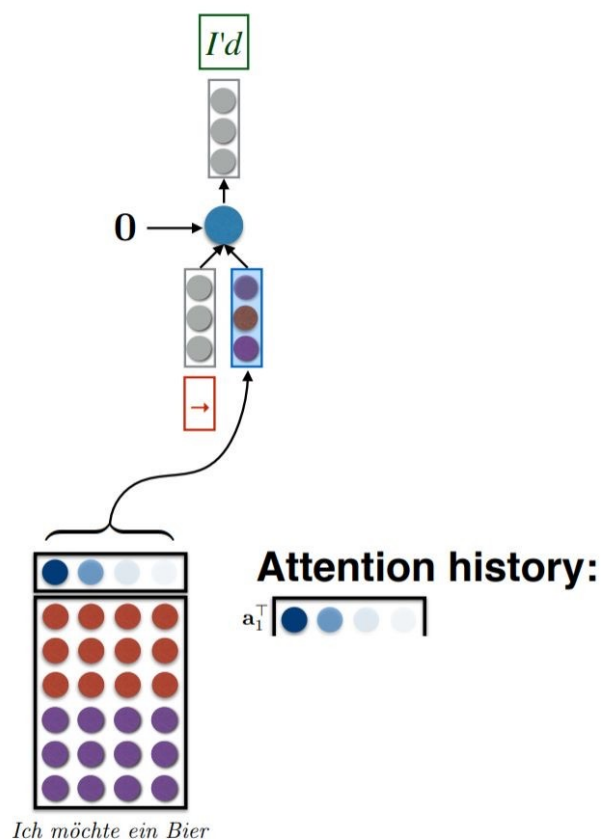


Ich möchte ein Bier

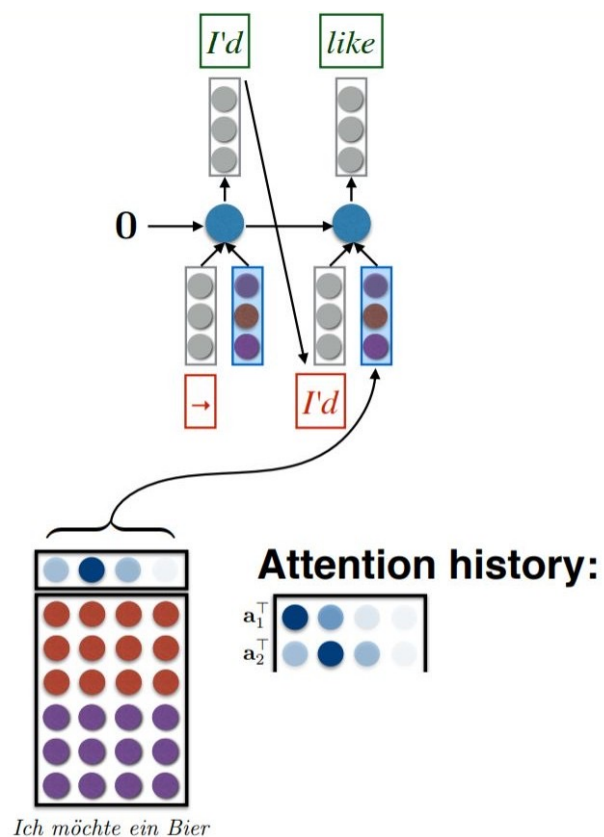
解码器：循环神经网络+注意力机制



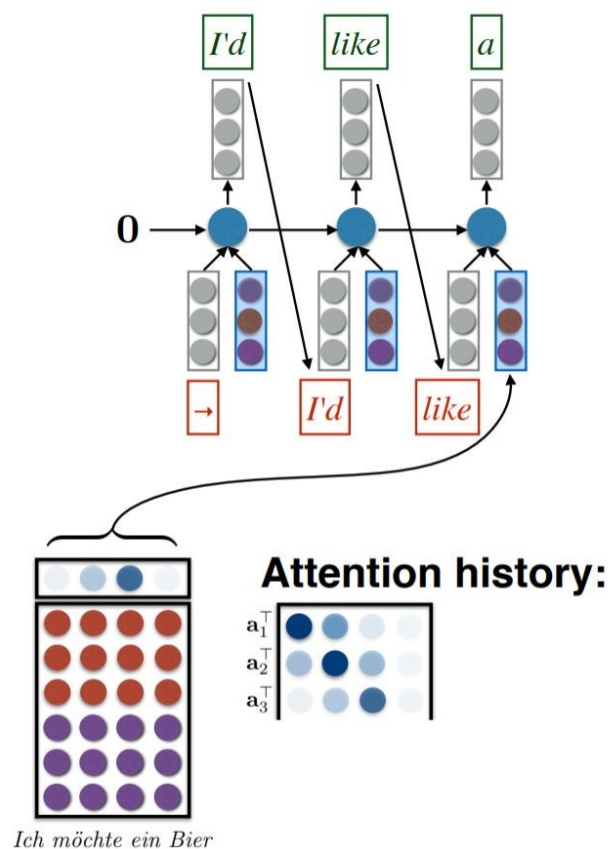
解码器：循环神经网络+注意力机制



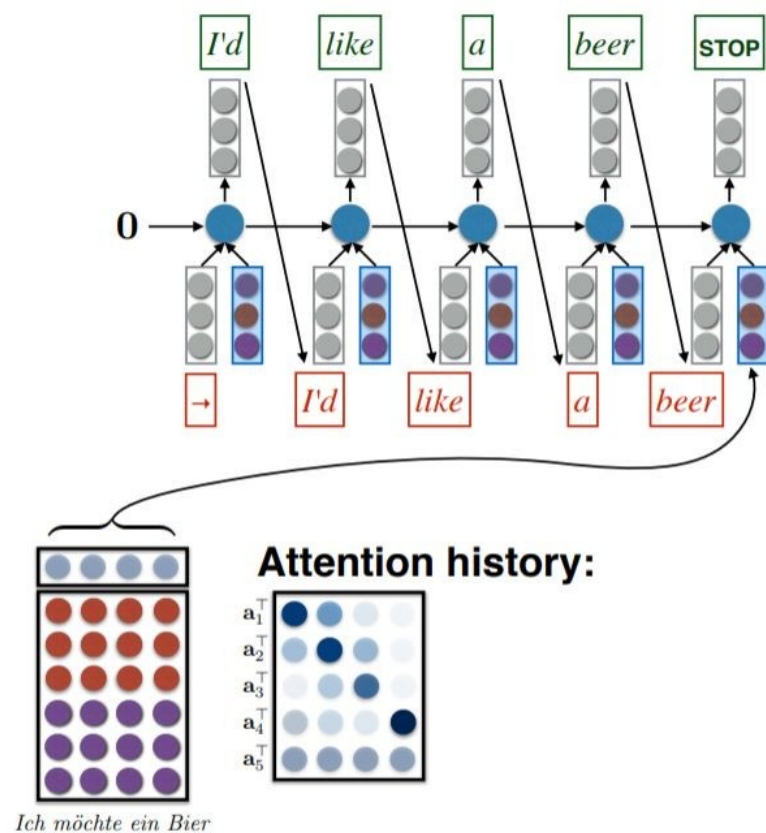
解码器：循环神经网络+注意力机制



解码器：循环神经网络+注意力机制



解码器：循环神经网络+注意力机制



大纲

- 前言
- 循环神经网络
- 训练的常见问题
- 循环神经网络的使用
- **循环神经网络的理解**
- 循环神经网络变种

LSTMs学到什么？

Andrej Karpathy, Justin Johnson, Li Fei-Fei: Visualizing and Understanding Recurrent Networks. CoRR abs/1506.02078 (2015)

- 加法连接机制让单个神经元节点具备了出人意料的可解释性
- 对应tanh(c)取值
 - -1: 红色
 - +1: 蓝色

Cell sensitive to position in line:

```
The sole importance of the crossing of the Berezina lies in the fact that it plainly and indubitably proved the fallacy of all the plans for cutting off the enemy's retreat and the soundness of the only possible line of action--the one Kutuzov and the general mass of the army demanded--namely, simply to follow the enemy up. The French crowd fled at a continually increasing speed and all its energy was directed to reaching its goal. It fled like a wounded animal and it was impossible to block its path. This was shown not so much by the arrangements it made for crossing as by what took place at the bridges. When the bridges broke down, unarmed soldiers, people from Moscow and women with children who were with the French transport, all--carried on by vis inertiae--pressed forward into boats and into the ice-covered water and did not, surrender.
```

Cell that turns on inside quotes:

```
"You mean to imply that I have nothing to eat out of.... On the contrary, I can supply you with everything even if you want to give dinner parties," warmly replied Chichagov, who tried by every word he spoke to prove his own rectitude and therefore imagined Kutuzov to be animated by the same desire.
```

```
Kutuzov, shrugging his shoulders, replied with his subtle penetrating smile: "I meant merely to say what I said."
```

Cell that robustly activates inside if statements:

```
static int __dequeue_signal(struct sigpending *pending, sigset_t *mask, siginfo_t *info)
{
    int sig = next_signal(pending, mask);
    if (sig) {
        if (current->notifier) {
            if (sigismember(current->notifier_mask, sig)) {
                if (!current->notifier)(current->notifier_data) {
                    clear_thread_flag(TIF_SIGPENDING);
                    return 0;
                }
            }
        }
        collect_signal(sig, pending, info);
    }
    return sig;
}
```

A large portion of cells are not easily interpretable. Here is a typical example:

```
/* Unpack a filter field's string representation from user-space
 * buffer. */
char *audit_unpack_string(void **bufp, size_t *remain, size_t len)
{
    char *str;
    if (!*bufp || (len == 0) || (len > *remain))
        return ERR_PTR(-EINVAL);
    /* Of the currently implemented string fields, PATH_MAX
     * defines the longest valid length.
     */
```

Cell that turns on inside comments and quotes:

```
/* Duplicate LSM field information. The lsm_rule is opaque, so
 * re-initialized. */
static inline int audit_dupe_lsm_field(struct audit_field *df,
                                     struct audit_field *sf)
{
    int ret = 0;
    char *lsm_str;
    /* Our own copy of lsm_str */
    lsm_str = kstrdup(sf->lsm_str, GFP_KERNEL);
    if (unlikely(!lsm_str))
        return -ENOMEM;
    df->lsm_str = lsm_str;
    /* Our own (refreshed) copy of lsm_rule */
    ret = security_audit_rule_init(df->type, df->op, df->lsm_str,
                                   (void **)&df->lsm_rule);
    /* Keep currently invalid fields around in case they
     * become valid after a policy reload. */
    if (ret == -EINVAL) {
        pr_warn("audit rule for LSM '%s' is invalid\n",
                df->lsm_str);
        ret = 0;
    }
    return ret;
}
```

Cell that is sensitive to the depth of an expression:

```
#ifdef CONFIG_AUDITSYSCALL
static inline int audit_match_class_bits(int class, u32 *mask)
{
    int i;
    if (classes[class]) {
        for (i = 0; i < AUDIT_BITMASK_SIZE; i++)
            if (mask[i] & classes[class][i])
                return 0;
    }
    return 1;
}
```

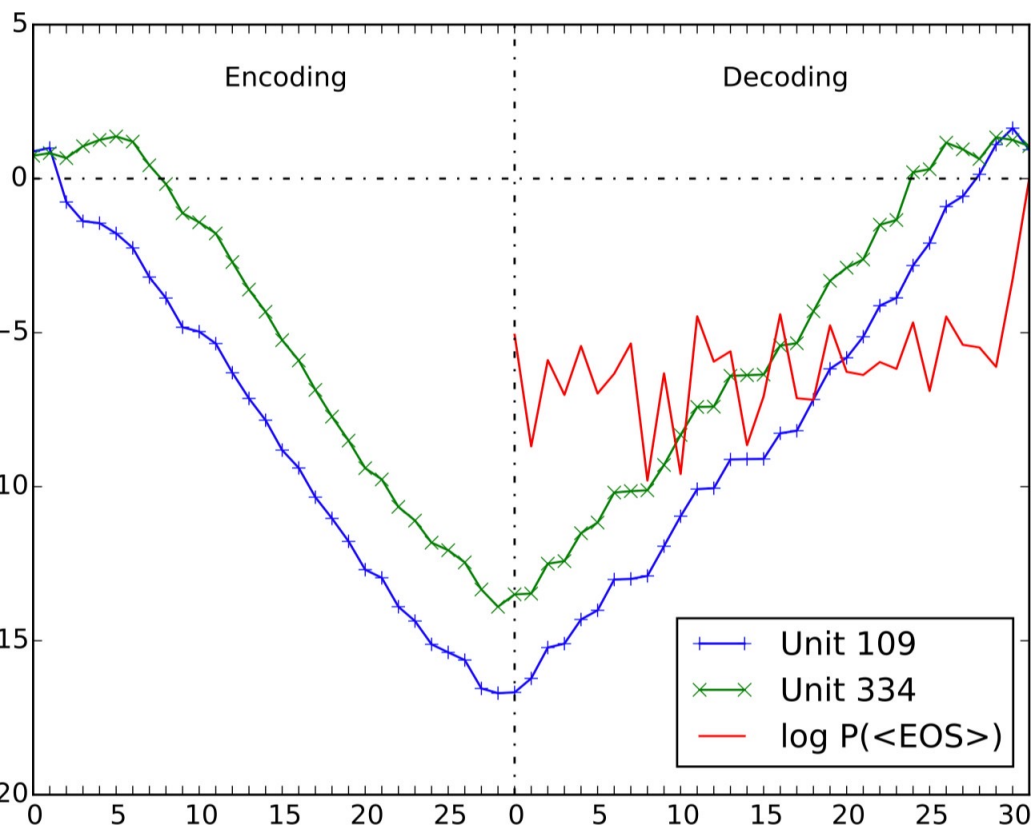
Cell that might be helpful in predicting a new line. Note that it only turns on for some "":

```
char *audit_unpack_string(void **bufp, size_t *remain, size_t len)
{
    char *str;
    if (!*bufp || (len == 0) || (len > *remain))
        return ERR_PTR(-EINVAL);
    /* Of the currently implemented string fields, PATH_MAX
     * defines the longest valid length.
     */
    if (len > PATH_MAX)
        return ERR_PTR(-ENAMETOOLONG);
    str = kmalloc(len + 1, GFP_KERNEL);
    if (unlikely(!str))
        return ERR_PTR(-ENOMEM);
    memcpy(str, *bufp, len);
    str[len] = 0;
    *bufp += len;
    *remain -= len;
    return str;
}
```


LSTMs学到什么?

分析情感倾向

统计句子长度



Shi et al. : Why Neural Translations are the Right Length. EMNLP 2016

12/19/25

25 August 2003 League of Extraordinary Gentlemen: Sean Connery is one of the all time greats and I have been a fan of his since the 1950's. I went to this movie because Sean Connery was the main actor. I had not read reviews or had any prior knowledge of the movie. The movie surprised me quite a bit. The scenery and sights were spectacular, but the plot was unreal to the point of being ridiculous. In my mind this was not one of his better movies it could be the worst. Why he chose to be in this movie is a mystery. For me, going to this movie was a waste of my time. I will continue to go to his movies and add his movies to my video collection. But I can't see wasting money to put this movie in my collection

I found this to be a charming adaptation, very lively and full of fun. With the exception of a couple of major errors, the cast is wonderful. I have to echo some of the earlier comments -- Chynna Phillips is horribly miscast as a teenager. At 27, she's just too old (and, yes, it DOES show), and lacks the singing "chops" for Broadway-style music. Vanessa Williams is a decent-enough singer and, for a non-dancer, she's adequate. However, she is NOT Latina, and her character definitely is. She's also very STRIDENT throughout, which gets tiresome. The girls of Sweet Apple's Conrad Birdie fan club really sparkle -- with special kudos to Brigitta Dau and Chiara Zanni. I also enjoyed Tyne Daly's performance, though I'm not generally a fan of her work. Finally, the dancing Shriners are a riot, especially the dorky three in the bar. The movie is suitable for the whole family, and I highly recommend it.

Alec Radford, Rafal Józefowicz, Ilya Sutskever: Learning to Generate Reviews and Discovering Sentiment. CoRR abs/1704.01444 (2017)

IB00383-自然语言处理

64

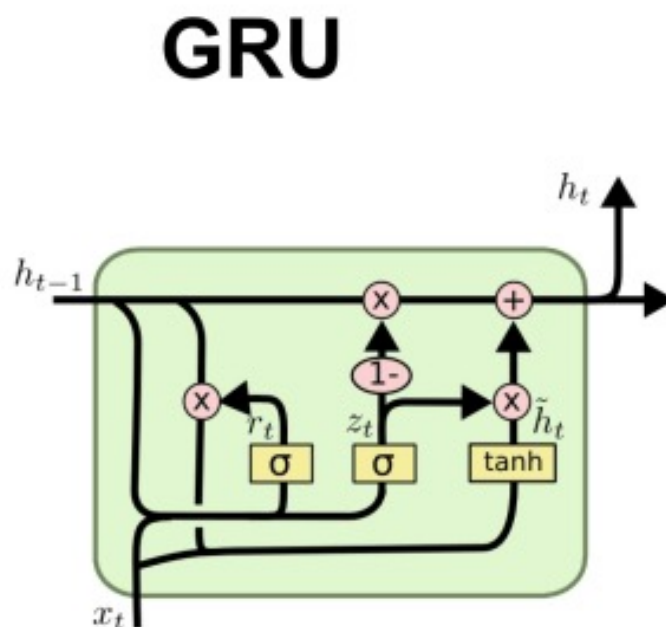
大纲

- 前言
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门控循环单元

Gated Recurrent Unit, GRU

- Cho et al. 2014
- 与LSTM不同之处
 - 少一个门：
 - 更新门
 - 重置门
- 优势
 - 训练速度比LSTM快
 - 效果跟LSTM差不多



Update gate: controls what parts of hidden state are updated vs preserved

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

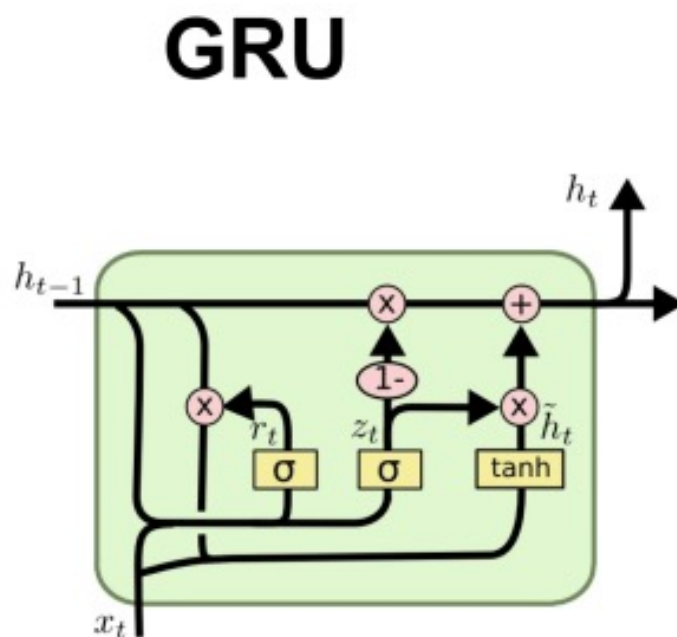
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

<https://clay-atlas.com/blog/2020/06/02/machine-learning-cn-gru-note/>

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Reset gate: controls what parts of previous hidden state are used to compute new content.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

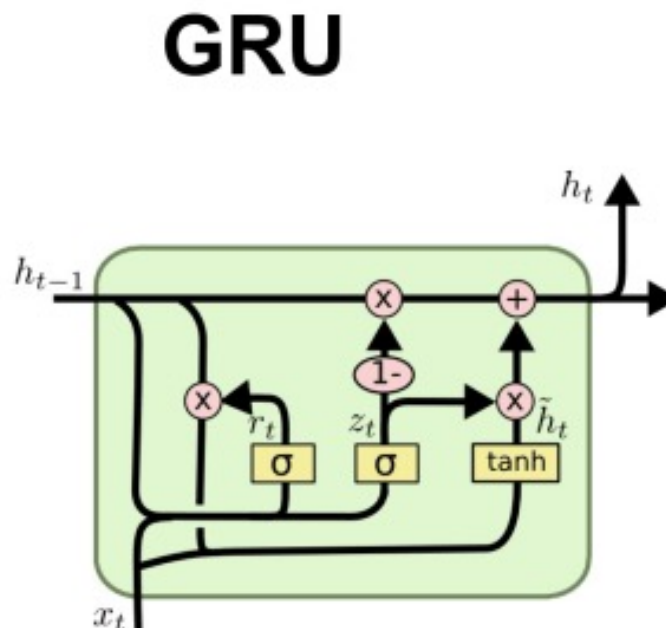
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

<https://clay-atlas.com/blog/2020/06/02/machine-learning-cn-gru-note/>

门控循环单元

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New hidden state content: reset gate selects useful parts of prev hidden state. Use this and current input to compute new hidden content.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

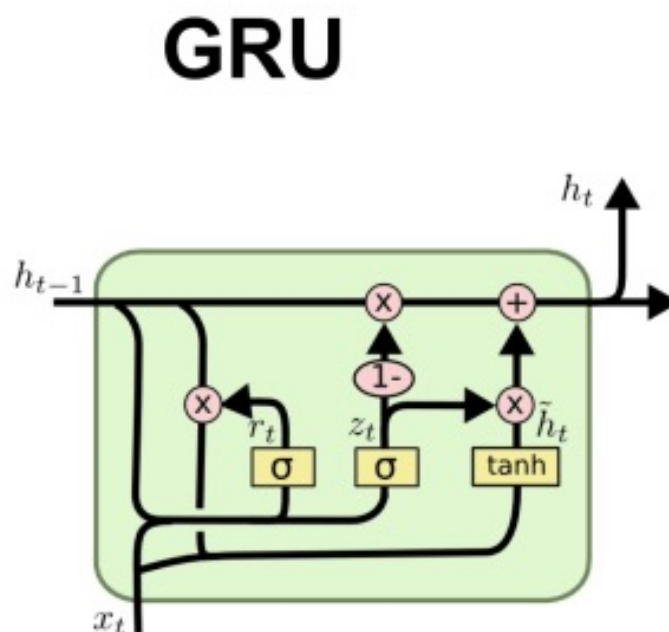
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

<https://clay-atlas.com/blog/2020/06/02/machine-learning-cn-gru-note/>

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Hidden state: update gate simultaneously controls what is kept from previous hidden state, and what is updated to new hidden state content.

$$z_t = \sigma(W_z \cdot [h_{t-1}, x_t])$$

$$r_t = \sigma(W_r \cdot [h_{t-1}, x_t])$$

$$\tilde{h}_t = \tanh(W \cdot [r_t * h_{t-1}, x_t])$$

$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$

<https://clay-atlas.com/blog/2020/06/02/machine-learning-cn-gru-note/>

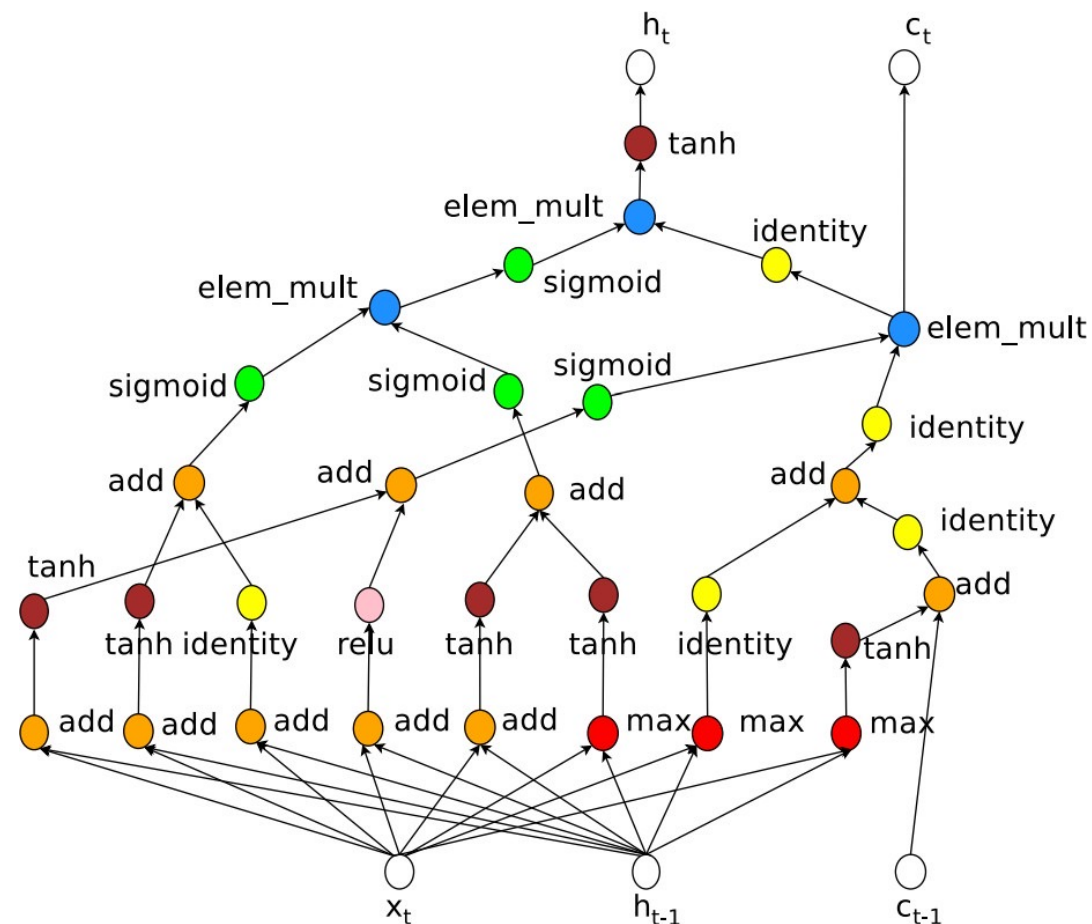
Neural Architecture Search for LSTMs

(Greff et al. 2015, Zoph and Le 2017)

- 大量测试多种不同类型的网络架构, 包括人工设计与自动搜索两类方式

结论

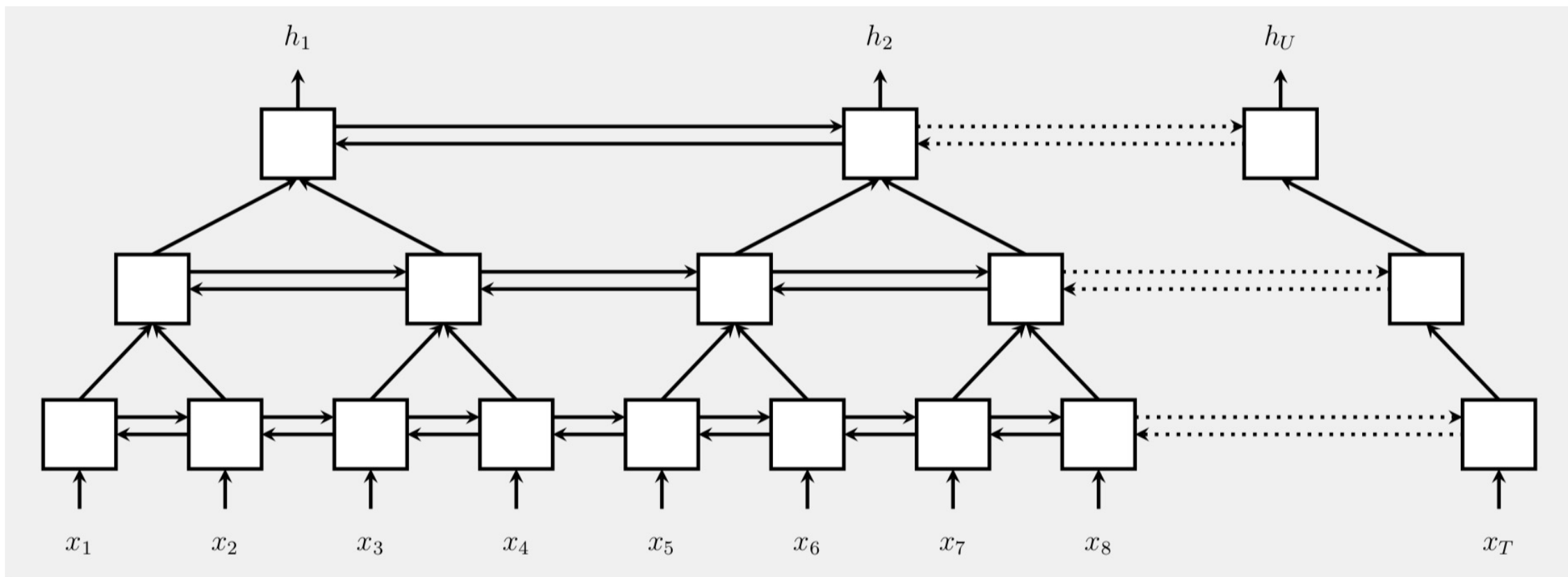
- 基础LSTM表现不错
- 其他变体(如耦合输入门/遗忘门的结构)的效果亦可观



Multi-scale/Pyramidal RNNs

(e.g. Chan et al. 2015)

- 构建不同规模的多循环神经网络



一句话总结

- 循环神经网络

- 定义
- 训练
- 使用
- 理解
- 变种

- 课外阅读

- [SLP3] Ch. 8, 17

